MATH INQUIRERS HOLIDAY PROGRAM

Math Inquirers Level 1 Tutor Guide

Holiday program for Level 1

Notice it. Test it. Explain why.

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Program guide for tutors: schedule, lesson routine, ten minute-by-minute lesson scripts with answer keys, a progress tracker and a parent summary. Math Inquirers is online first: every lesson runs live on video; the face-to-face notes are kept for partner centres.

The program at a glance

Level	Level 1 (ages 6 to 7), for children ready for more thinking than standard practice
Length	10 lessons of 90 minutes, two a week over the 5 weeks of the holidays
Group size	Up to 8 children per tutor, live online. Pilot with 6. Add a second tutor or helper above 8.
Focus	Thinking skills: spotting patterns, stating rules, working backwards, reasoning and explaining. Not speed drills.
The arc of every lesson	Start concrete with objects, record in a table, spot the pattern, state the rule, use it (often backwards), then explain why.
What children take home	A student workbook with an inquiry journal, 10 worksheets and a certificate
Before tutors start	Every tutor signs the tutor confidentiality agreement before receiving any materials.

Schedule

Week	Lesson	Title	Topic	Big idea
Week 1a	1	**Count with the Frog**	Number Patterns	Pattern spotting
Week 1b	2	**Number Bond Inquiry**	Number Bonds	Structure of numbers

Week 2a	3	**Let's Plan the Fruit Party**	Picture Graphs	Data reading
Week 2b	4	**Pattern Inquiry**	Repeating and Growing Patterns	Core identification
Week 3a	5	**Mystery Number**	Working Backwards	Inverse operations
Week 3b	6	**Coin Combos**	Money	Systematic listing
Week 4a	7	**Shape Safari**	2D Shapes	Classifying
Week 4b	8	**Logic Lab**	Logic and Sorting	Deduction
Week 5a	9	**Measure It!**	Length	Comparing
Week 5b	10	**Challenge Day**	Mixed Puzzles	Applying and connecting ideas

Suggested pairing: two lessons a week, for example Tuesday and Thursday, or Saturday and Sunday. Keep the same time slot all five weeks.

The 90-minute lesson routine

Every lesson uses the same routine so children know what to expect and tutors can prepare quickly.

Minutes	Part	What happens	Face to face	Online
0–10	**Warm-up**	A quick game (0 to 5) and quick questions (5 to 7). The last 3 minutes (7 to 10) are the sneak-peek puzzles: two short puzzles on ideas we have not learnt yet. The first is one that any child can try with objects, a drawing or counting. The second makes you think. Answers are never put on boards in front of the group.	Sit in a circle or horseshoe.	Cameras on. Answers by thumbs, hands or whisper. Sneak-peek answers are shown to the tutor only, on camera.
10–25	**Explore**	Hands-on in pairs. Objects first, no writing yet. Any pair may also choose the “Go further” what-if question on the slide from minute 15, so nobody waits.	Materials box on each table.	Each child has their own materials at home. Pairs go to breakout rooms.
25–35	**Mini-lesson**	Tutor builds the table on the board, names the rule and asks what stays the same and what changes.	Whiteboard.	Shared slide and annotation tool.
35–40	**Brain break**	Stand, stretch, drink water.	A short movement game.	Stand up on camera. Count the stretches together.

40–72	**Worksheet**	Tutor reads questions 1 and 2 aloud, then pairs read the rest. Start in pairs, finish alone, then check together (the last 4 minutes, starting with a 1-minute stand-up stretch). Tutor asks “how do you know?” rather than giving answers. Every child’s workbook says: “When you finish this page you may choose: carry on, or try Think harder.” Think harder (T), Go further (C) and the Investigation are printed for every child. The tutor never announces who is on which page.	Printed workbook, tutor circulates.	Printed workbook. Child holds the page up to the camera. Tutor checks one or two answers each.
72–80	**Talk time**	Volunteers share one answer and explain it; passing is always fine. The tutor asks for volunteers first and never calls a child by name to answer. No tallies, no rankings.	Volunteers first. A child who prefers can show a drawing or pass.	Volunteers unmute. Others may pass or type in chat.
80–90	**Exit ticket, look-back and stamp**	One question on a slip of paper (80 to 88). Stamp the inquiry journal. Then a two-minute look-back (88 to 90) in the workbook. A pass is always allowed.	Paper slip, tutor stamps.	Child writes the answer and shows the camera. Tutor stamps the next time or posts it.

Teaching habits that matter

- Ask “How do you know?” after every answer, right or wrong.
- Wait 10 seconds before helping. Let children struggle a little.
- Praise the thinking (“You checked it two ways”), not just the answer.
- Let children disagree with each other and settle it with objects or a drawing.
- If a child finishes early, use the Challenge or Extension question, not extra drill.

Sneak-peek puzzles, choice time and look-back (every lesson)

- Sneak-peek puzzles (minutes 7 to 10). Two questions on the lesson’s key idea. The first is an access question: every child can try it with objects, drawing or counting, so everyone starts with a win. The second makes you think. Say: “These are ideas we have not learnt yet. A guess is a great start, and ‘I do not know’ is welcome. Nobody is marked.” Children write or draw on their own paper. Answers are not shown on boards in front of the group: face to face, each child holds their own paper low so that only

the tutor sees it; online, each child writes on their own paper and shows it to the tutor on camera, or everyone holds their paper to the camera at the same moment. Never use boards held up in front of the group. Never write scores down or announce them.

- Choice time. Every workbook has the same neutral line on page 1: “When you finish this page you may choose: carry on, or try Think harder.” Any child may choose. There is no list of who is ahead, and no criterion is printed. The tutor never announces who is on which page, and never names a child in front of the group. The tutor invites the WHOLE group at once, so nobody is singled out, for example: “When you have finished Start here, you can choose Think harder, Go further or the Investigation.” Say the same to any child who asks. Pairs are not split unless both children agree. A child who finds Think harder too hard goes back to Start here, with no fuss.
- Online with a group of 8, the tutor can see each child’s page because each child already holds it up to the camera. Above 8 children, the helper keeps an eye on the children who have moved on while the tutor works with the others.
- Talk time. Ask for volunteers first. Never call a child by name to answer. A child may pass, show a drawing or let a partner read for them. No public tallies and no rankings: do not say who is first or who got most right.
- Look-back (minutes 88 to 90, two minutes). The box is on the Choose and look-back page of the workbook, which every child reaches: “Something I did well”, “The hardest part was …”, “What helped me”, with tick options that include “I asked for help” and “I am still stuck (my tutor will look at this after class)”. Passing is always allowed. The box is never collected, shown or read out in front of the group. The tutor may look at it afterwards. If a child ticks “I am still stuck”, the tutor looks at the box after class and checks in at the start of the next lesson, in the open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy (“Drawing it got you unstuck”), not the speed.
- The Investigation is printed on the Choose and look-back page for every child, with an easy first step, and is optional. It has several valid answers. The key gives a model answer and what to look for. Accept any correct noticing, and ask “How do you know?”.

Setting up face to face

- Seat children in pairs. Keep a materials box per pair so no time is lost.
- Print the workbook double-sided and staple it. Print the cut-out page on card for Lessons 6 and 9.
- Keep the answer key with the tutor only. Collect worksheets at the end unless the group is told to keep them.
- Use a whiteboard or large paper for the mini-lesson table.

Setting up online

- Send parents the workbook and materials list one week before. Needed: counters or coins, two colours of paper pieces, a cup, straws or lolly sticks (never real matches or toothpicks), paper, a pencil and a cm ruler (Lesson 9).
- Use one approved platform with breakout rooms and screen share. Admit only registered children and never share the link publicly.
- Ask the parent to sit nearby but not give answers. The camera should show the child’s page and hands.

- Hold up the answer page to the camera at the end of each lesson, or the parent takes a photo and sends it to the tutor through the approved parent channel (the tutor never asks a child to send anything).
- Do not record lessons (this is in the tutor agreement). Add a second tutor or helper above 8 children.

Materials master list

Lesson	Title	Face to face	Online set-up
1	Count with the Frog	Paper number strip 0 to 50 (printed in the Inquirer Kit at the back of the workbook; or floor tape), counters, two crayons.	Slide number line the tutor can mark on; each child uses a paper strip and a pencil.
2	Number Bond Inquiry	Ten-frames, red and yellow counters, a cup, a paper triangle with 6 circles.	Slide ten-frame with a hidden-counter reveal; each child uses 10 coins or paper pieces of two colours.
3	Let's Plan the Fruit Party	Sticky dots, 5 fruit pictures on the wall, counters.	Poll on the slide (thumbs up, no typing needed) and a shared graph the tutor builds live.
4	Pattern Inquiry	21 straws, lolly sticks or paper strips per pair (never real matches), shape cut-outs.	Drawing on paper or screen; children use straws, lolly sticks, paper strips or pencils at home (never real matches or toothpicks).
5	Mystery Number	Paper cups, counters, number machine cards.	Slide with a cup that hides counters; the tutor reveals the total and the children work back.
6	Coin Combos	Play coins (5¢, 10¢, 20¢, 50¢, $1), about 6 of each per pair, play notes ($2, $5, $10, $50), mini whiteboards.	Print the cut-out coins and notes at the back of the workbook, or use real coins and notes at home.
7	Shape Safari	Paper squares, scissors, pattern blocks, 2 shape riddles on cards.	Each child has a paper square to fold and cut on camera; shape hunt at home.
8	Logic Lab	Two loops of string, buttons or small objects (small objects only with a grown-up nearby), clue cards.	Children sort objects from home by a rule shown on the slide and explain the rule on camera. Buttons and other small objects: only with a grown-up nearby.
9	Measure It!	Linking cubes or paper cube strips, paper clips, erasers, objects to measure, a cm ruler for each child, mini whiteboards.	Print the cube strip at the back of the workbook; measure household objects. Each child needs a cm ruler.

10	Challenge Day	Four puzzle cards (coins, patterns, logic, measuring; printed in the workbook and shown on the Lesson 10 slide), play coins, cubes, certificates, stamps, mini whiteboards.	The four cards are printed in the workbook and shown on the slide. Each pair chooses one card. Pairs swap at minute 20. Reporting back is optional: say one thing you tried, not the answer.

How to use the scripts

Every lesson below is a minute-by-minute script. The slides match it: one slide or two for each part. You do not have to say the words exactly, but keep the questions, because the questions are what makes children think.

SAY	Words to say aloud, in plain language for 6- and 7-year-olds. Said in your own voice, not read from the page.
DO	What you do: set up, hand out, show a slide, or move around the room.
ASK / EXPECT	Questions to put to the group, with the answer you are listening for. Let children answer before you explain.
IF STUCK	A ladder of help. Give hint 1, wait ten seconds, then give hint 2. Do not give the answer.

Five habits that run through every script

- Predict first. Before children test anything, ask for a guess and write it down. A wrong guess is useful because we can test it.
- Wait ten seconds after every question. Count silently. Most children answer in the last four.
- Ask “How do you know?” after right answers too, so that children do not read your face for the answer.
- Praise the thinking, not the answer: “You checked it two ways” beats “Well done”.
- When an answer is wrong, say “Let us test that,” and use objects, not “No”.

Answering a wrong answer

If...	Then...
A child gives a wrong answer	“Thank you. Let us test it with the counters.” Then build it together.
A child gives a right answer but cannot explain	“Show me with your counters.” Then: “Tell your partner how you know.”
A child says “I do not know”	“What do you notice? What could you try first?” Offer hint 1.
Two children disagree	“Both of you, convince the group using objects.” Let the objects decide.

Lesson 1: Count with the Frog

Week 1 (first lesson) • Number Patterns • Workbook: Lesson 1

Objective	Skip count in 2s, 3s and 5s; find and state an add or take-away rule; notice numbers that both counting patterns share.
Thinking skills	Pattern spotting, rule finding, generalising
Materials	Paper number strip 0 to 50 (printed in the Inquirer Kit at the back of the workbook; or floor tape), counters, two crayons.
Online set-up	Slide number line the tutor can mark on; each child uses a paper strip and a pencil.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Good morning, inquirers! Today we are frogs on a very long number strip."*

"A frog never jumps at random. Its jumps follow a rule. Our job is to notice the rule, test it with a counter, check our answer and explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–7 min Warm-up game and quick questions

DO Skip-count stomp. Stand up! Count in 2s from 0 to 20, stomping on each number in the pattern and whispering the rest. Then count in 5s to 50, and in 3s to 30.

SAY *"Everybody stand up. When I say a number that is in our pattern, STOMP. When it is not, whisper it."*

"Ready? We count in 2s. Zero, two, four..."

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Count in 2s: 2, 4, 6, __	8
Count in 5s: 5, 10, __, 20	15
Count in 3s: 3, 6, 9, __	12
Count back in 2s: 20, 18, __	16
After 4 jumps, which goes further: jumps of 5 or jumps of 3?	5s reach 20; 3s reach 12

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): A frog hops 3 each time from 0: 3, 6, 9 … Use a counter or draw the hops. What number does it land on next?	12. Hop on 3 more from 9: 10, 11, 12.
Sneak-peek 2 (thinking): A bunny starts at 1 and hops 3 each time: 1, 4, 7 … Will it land on 20? How do you know?	No. It lands on 19, then 22, so it jumps over 20. (Keep hopping, or notice it only lands on numbers that are 1 more than a 3s number.)

10–25 min Explore (in pairs, with objects)

SAY *"Before we hop, I want a prediction. Frog 1 jumps in 4s. Frog 2 jumps in 5s. Will they ever land on the same number?"*

"Thumbs up for yes, sideways for maybe, down for no. Now write your prediction at the top of your strip."

DO 1. Make a number strip from 0 to 50.

2. Frog 1 starts on 0 and jumps in 4s. Mark each landing with a red cross.

3. Frog 2 starts on 0 and jumps in 5s. Mark each landing with a blue circle.

4. Write down the numbers that have BOTH a cross and a circle.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck on the strip: put your counter on 0 and say "four, eight" out loud as you hop.

Losing track: write each landing under the strip: 4, 8, 12 …

Finished early: ask "Frog 3 jumps in 10s. Where will it meet frog 1?"

DO At about minute 22, say “Pencils down” and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Which numbers did both frogs land on?	20 and 40 (and 0).
What do you notice about them?	They are in the 20 times table: 5 jumps of 4 or 4 jumps of 5.
What if frog 2 jumped in 10s instead?	Both frogs still land on 20 and 40.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *“Look at the table. The numbers are 2, 8, 14, 20. I notice they go up. How much from 2 to 8? Six. From 8 to 14? Six. From 14 to 20? Six. I check every pair of neighbours and it is always six.”*

“So my rule is add 6. To find the 5th number I add 6 to the 4th: 20 + 6 = 26.”

“Now look at the numbers that go down. If they get smaller, my rule is take away. Always check: does my rule work on the last pair too?”

Worked example on the slides	Answer
Build the table. Start at 2. What do you add each time? Rule: add 6. What is the 5th number?	26. Rule: add 6.
Going down. The numbers get smaller, so the rule is take away. What is the rule? What comes next?	Take away 4. Next is 24.

SAY *“The key idea today is: find the rule. The rule tells you how much the number changes each time. Say it with me.”*

DO Show the key idea slide at minutes 33 to 35. The “common mistakes” on that slide are for you, not the children. Watch for them during the worksheet:

- Counting the numbers in between instead of the hops (4 to 7 counted as 2).
- Writing "add 3" when the numbers go down (take away 3).
- Restarting the count instead of adding to the last number.

35–40 min Brain break

DO Frog jumps! Do 10 star jumps counting in 2s out loud, then 5 hops counting in 5s.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Julia Robinson (USA, 1919-1985) loved the counting numbers, and as a small child she liked arranging pebbles. She grew up to be a mathematician who worked on hard number puzzles. Link to our lesson: Julia loved counting numbers and arranging objects, and today the children hop along the number strip and find the rule that makes a counting pattern. Then offer

the try-it question, only if you like: “Julia puts pebbles in rows. Row 1 has 3 pebbles, row 2 has 6, row 3 has 9. How many pebbles are in row 5? What is the rule?” (the answer is in the answer key). SKIP: if time is short or the group is tired, say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 1 aloud and read questions 1 and 2 aloud. Say “Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder.”

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Point to a number and ask: “How do you know this number is in the pattern?”
- Ask: “What is the jump size? Show me on the strip.”
- Ask: “Check your rule on the last pair as well. Does it still work?”

IF STUCK Hint 1: “Look at two neighbours. How far apart are they?”

Hint 2: “Hop your counter and count each hop out loud.”

Hint 3: do the first one together, then let the child do the next one alone.

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: Fill in the missing numbers: 3, 8, __, 18, __, 28	13 and 23. Add 5.
Workbook Think harder T2: Fill in the missing numbers: 40, __, 28, __, 16, 10	34 and 22. Take away 6 (use the two neighbours 16 and 10).
Workbook Think harder T3: Is 30 in the pattern 4, 8, 12, 16 … ? How do you know?	No. The pattern goes 28, 32, so it skips 30.
Workbook Think harder T4: Ali counts in 2s from 2 up to 50. Mei counts in 5s from 5 up to 50. Which numbers do both of them say? What do you notice?	10, 20, 30, 40 and 50. They all end in 0.

Workbook Go further C1: 1, 2, 4, 7, 11, 16 … What is the 8th number? Explain how the pattern grows.	29. The gaps are 1, 2, 3, 4, 5, 6, 7, so the 7th is 22 and the 8th is 29.
Workbook Go further C2: Ali counts up from 2 in 3s. Mei counts up from 1 in 4s. Which numbers up to 50 are in both patterns? What do you notice about the gaps between them?	5, 17, 29 and 41. Each one is 12 more than the one before.
Workbook Go further C3: Make up your own two counting patterns: choose a start and a jump for each, so that they share some numbers up to 50. Write both patterns and ring the shared numbers. What do you notice about the gaps between the shared numbers?	Open. Check by writing both lists and ringing the matches. Example: start 2 add 4 (2, 6, 10 …) and start 3 add 3 (3, 6, 9 …) share 6, 18, 30 and 42, with a gap of 12 each time.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: "Who would like to share?" Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *"Who would like to tell us numbers that both frogs landed on?"*

"Why do the numbers both frogs share all end in 0?"

Discussion question	Listen for
Why do the numbers both frogs land on all end in 0?	Jumps of 5 land on numbers ending in 0 or 5. Jumps of 4 always land on even numbers. A number that is both must end in 0.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Next two numbers: 7, 10, 13, 16, __, __. What is the rule?	19, 22. Add 3.

SAY *"Today you noticed a rule, tested it with a counter and explained why. That is what an inquirer does."*

"Take your inquiry journal and I will stamp it when you hand me your exit ticket."

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Pupils invent a rule (start at 1, add 6) and swap with a partner to find it.

Answer key (workbook)

Q1 (a) 8, 10, 14 (b) 15, 25, 35 (c) 9, 15

Q2 (a) 16, 19; add 3 (b) 18, 15; take away 3 (c) 18, 22; add 4

Q3 Frog 1 lands on 4, 8, 12, 16, 20. Frog 2 lands on 5, 10, 15, 20. Both: 20.

Q4 (a) 10, 20, 30 (b) They all end in 0. We say them when we count in 10s. (Up to 50 the answer is 10, 20, 30, 40, 50: see Think harder T4.)

Q5 Stickers: 2, 5, 8, 11, 14, 17, 20, 23, 26, 29. (a) Box 10 has 29. (b) Even, odd, even, odd: they alternate.

Think harder T1 to T4 T1: 13 and 23. Add 5. T2: 34 and 22. Take away 6 (use the two neighbours 16 and 10). T3: No. The pattern goes 28, 32, so it skips 30. T4: 10, 20, 30, 40 and 50. They all end in 0.

Think harder slide 1) Yes to 39, no to 40 (3, 7, 11, 15, 19, 23, 27, 31, 35, 39, 43: it lands on 39 and jumps over 40). 2) 22 does not belong. It should be 20 (the pattern adds 5). 3) Add 4: from the 3rd to the 5th number is two equal jumps that make 8, and 4 + 4 = 8. The 2nd number is 11 − 4 = 7 and the 1st is 7 − 4 = 3. Check: 3, 7, 11, 15, 19.

Sneak-peek 1 12. Hop on 3 more from 9: 10, 11, 12.

Sneak-peek 2 No. It lands on 19, then 22, so it jumps over 20. (Keep hopping, or notice it only lands on numbers that are 1 more than a 3s number.)

Investigation (What if we change the hops?) Both land on: with 3s 6, 12, 18, 24, 30; with 4s 4, 8, 12, 16, 20, 24, 28 (every landing of frog 2, because 4 is a number in the 2s); with 5s 10, 20, 30; with 6s 6, 12, 18, 24, 30. Stays the same: 0 is always shared, and the shared numbers always form a pattern with equal jumps. Changes: the size of that jump (6, 4, 10, 6) and how many numbers are shared. Look for: both lists written out, a systematic check, and a reason why the 4s frog shares every landing (4 = 2 + 2). Accept any correct noticing.

Story break try-it (Julia Robinson) 15 pebbles. The rule is add 3 each time (row 4 has 12, row 5 has 15).

Lesson 2: Number Bond Inquiry

Week 1 (second lesson) • Number Bonds • Workbook: Lesson 2

Objective	Know the bonds of 10, use them to find bonds of 20, balance equations, and reason about missing numbers.
Thinking skills	Structure of numbers, systematic listing, reasoning about equality, guess and check
Materials	Ten-frames, red and yellow counters, a cup, a paper triangle with 6 circles.
Online set-up	Slide ten-frame with a hidden-counter reveal; each child uses 10 coins or paper pieces of two colours.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Today we are going to find every pair of numbers that joins to make 10. These pairs are called number bonds."*

"Then we will use them to make 20. And we will find out what the equals sign really means."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–7 min Warm-up game and quick questions

DO Hidden counters. Show 10 counters, then hide some under a cup. How many are under the cup? Repeat with new splits.

SAY *"I have 10 counters. Watch carefully. [Show all 10. Hide some under the cup.] There are 6 counters I can still see. How many are under the cup?"*

"How do you know? Say it as a number bond: "6 and __ make 10."

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
10 = 7 + __	3
10 = 2 + __	8
10 = __ + 5	5
10 = 10 + __	0
10 = __ + 1	9

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): There are 10 sweets. 7 are red and the rest are green. Use counters or draw them. How many are green?	3. 7 + 3 = 10.
Sneak-peek 2 (thinking): Ali has 7 red counters and 5 blue counters. Mei has 6 red counters and 6 blue counters. Who has more counters, or do they have the same? How do you know?	The same: 12 each. (7 + 5 = 12 and 6 + 6 = 12. Ali's 7 is 1 more than Mei's 6, but his 5 is 1 less than her 6.)

10–25 min Explore (in pairs, with objects)

SAY *"How many different ways do you think there are to make 10 with two numbers? Guess, and write it down."*

DO 1. Take a ten-frame and 10 counters of two colours.

2. Fill the ten-frame. Write the number of each colour, like 6 + 4.

3. Change some counters to make a new pair. Write it down.

4. Keep going until you think you have found them all.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start with all red counters. That is 10 + 0. Now change one to yellow.

Not sure if 3 + 7 and 7 + 3 are different: flip your ten-frame upside down and look.

Finished early: use your bonds of 10 to make bonds of 20 (16 + 4).

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many pairs did you find?	11 pairs from 0 + 10 to 10 + 0, or 6 different bonds if 3 + 7 and 7 + 3 are the same.
How do you know you have all of them?	List in order: the first number goes up by 1 and the second goes down by 1.
What happens to the second number when the first goes up by 1?	It goes down by 1.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Let me list them in order: 0 + 10, 1 + 9, 2 + 8. What do you notice? The first number goes up by 1 and the second goes down by 1. That is how I know I have them all."*

"Now 20. I know 6 + 4 = 10. So 16 + 4 = 20. The ones stay the same. Only the tens change."

"Look at 6 + 2 = 5 + __. Imagine a see-saw. Both sides must balance. 6 + 2 is 8, so the other side must be 8. What goes with 5? 3. Check: 5 + 3 = 8."

Worked example on the slides	Answer
Bonds of 10 help us make 20. Spot the link. What stays the same? What changes?	The ones digits are the same; the tens digit changes from 0 to 1.
What does "=" mean?. 6 + 2 = 5 + __ Both sides must be the same, like a balanced see-saw. What goes in the blank? How can you check?	3. Check: 6 + 2 = 8 and 5 + 3 = 8. Common mistake: writing 8 because the answer "comes next".

SAY *"The key idea today is: bonds and equals. Bonds of 10 are the building blocks for bonds of 20. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Reading "=" as "the answer comes next" (5 + 3 = 4 + __ answered 8).
- Counting 3+7 and 7+3 as different bonds when asked for different pairs.
- Magic triangle: trying numbers at random instead of looking at the corners.

35–40 min Brain break

DO Bond clap. The tutor says a number from 0 to 10; clap and say the partner that makes 10.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Robert Recorde (Wales, about 1510-1558) was a doctor who wrote maths books. In 1557 he used the sign = for the first time in a book, because no two things can be more

equal than two straight lines of the same length. Link to our lesson: Recorde's equals sign says both sides are worth the same, which is exactly what children check when they balance number bonds like 4 + 4 = 5 + 3. Then offer the try-it question, only if you like: “Use Recorde's sign. Fill in the missing number so that both sides are equal: 4 + 4 = 5 + __” (the answer is in the answer key). SKIP: if time is short or the group is tired, say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 2 aloud and read questions 1 and 2 aloud. Say “Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder.”

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Say 5 + 3 = 4 + __ like a see-saw. What does each side equal?”
- Magic triangle: “Which numbers are the corners? Each corner is counted in two sides.”
- Ask: “How do you know you have found all the pairs?”

IF STUCK Hint 1: “Work out the left side first. What is it?”

Hint 2: “Make the same amount with counters on both sides of the see-saw.”

Hint 3: “Try the number 4 in the blank. Is the right side the same?”

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: 5 + 4 = 6 + __	3
Workbook Think harder T2: 9 + 6 = __ + 10	5
Workbook Think harder T3: 6 + 4 = 10, so 16 + 4 = __ and 26 + 4 = __	20 and 30

Workbook Think harder T4: Find all the pairs of numbers from 1 to 11 that make 12. (3 + 9 and 9 + 3 count as the same pair.) How many pairs are there?	6 pairs: 1 + 11, 2 + 10, 3 + 9, 4 + 8, 5 + 7, 6 + 6. Going down one number at a time shows none are missed.
Workbook Go further C1: Put 1 to 6 in the magic triangle so each side adds to 11. What goes in the corners and middles?	Corners 2, 4, 6. Between 2 and 4: 5. Between 4 and 6: 1. Between 6 and 2: 3. (It is the only set of corners that works.)
Workbook Go further C2: Ali has 20 stickers. He keeps 9 and shares the rest between Ben and Mei. Ben gets 3 more than Mei. How many does each get?	Ben 7, Mei 4. (20 − 9 = 11 are shared; take away Ben's extra 3: 8, share 4 and 4; Ben has 4 + 3 = 7.)
Workbook Go further C3: Change the magic triangle: use the numbers 2, 3, 4, 5, 6 and 7 instead of 1 to 6. Can you make every side add to the same total? Try it. Which totals can you make?	Open. Yes, it still works. Totals 12, 13, 14 and 15 are possible. Example for 12: corners 2, 3, 4 with 7 between 2 and 3, 5 between 3 and 4 and 6 between 4 and 2.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: “Who would like to share?” Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *“Mei says 4 + 4 = 8 + 1. Who agrees with Mei? How can we check?”*

“What does the equals sign really mean?”

Discussion question	Listen for
Mei says 4 + 4 = 8 + 1. How do you check? What does "=" really mean?	4 + 4 = 8 but 8 + 1 = 9, so Mei is wrong. The equals sign means both sides have the same value, not “the answer comes next”.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
10 = 6 + __ 20 = 13 + __	4 and 7.

SAY *“You found every bond of 10 and you can say how you know you have them all. That is explaining why.”*

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Make every side of the magic triangle total 11 (one way: corners 2, 4, 6 with 5, 1, 3 between them). Then ask: can every side total 13? No: the biggest possible side total is 12, with 4, 5, 6 in the corners.

Answer key (workbook)

Q1 (a) 7 (b) 4 (c) 2 (d) 5 (e) 10

Q2 (a) 2+8, 3+7, 4+6, 5+5 (b) The second number goes down by 1 (c) 6 different bonds

Q3 (a) 20 and 20 (b) 20 and 20 (c) 8, 11, 5

Q4 (a) 4 (b) 4 (c) 6 (d) 2 (e) No. 4 + 4 = 8 but 8 + 1 = 9.

Q5 (a) Sides of 9: corners 1, 2, 3 (any order). Between 1 and 2: 6. Between 2 and 3: 4. Between 3 and 1: 5. (b) Sides of 10: corners 1, 3, 5. Between 1 and 3: 6. Between 3 and 5: 2. Between 5 and 1: 4.

Q6 4 and 6. Challenge: 4 and 8.

Think harder T1 to T4 T1: 3 T2: 5 T3: 20 and 30 T4: 6 pairs: 1 + 11, 2 + 10, 3 + 9, 4 + 8, 5 + 7, 6 + 6. Going down one number at a time shows none are missed.

Think harder slide 1) 10 and 16 (all three are 20). 2) 8 (20 − 4 = 16, and 8 + 8 = 16). 3) 6 pairs: 5 + 15, 6 + 14, 7 + 13, 8 + 12, 9 + 11, 10 + 10.

Sneak-peek 1 3. 7 + 3 = 10.

Sneak-peek 2 The same: 12 each. (7 + 5 = 12 and 6 + 6 = 12. Ali's 7 is 1 more than Mei's 6, but his 5 is 1 less than her 6.)

Investigation (Is it always true?) Yes. All 11 pairs from 0 + 10 to 10 + 0 are both odd or both even (1 + 9, 3 + 7 … and 0 + 10, 2 + 8 …); the same holds for all 21 pairs that make 20. Reason (accept child language): when one number goes up by 1 the other goes down by 1, so both change from odd to even together. Or: an odd and an even always make an odd total, never 10 or 20. Look for: testing every pair (not 2 or 3), writing them in order, and a reason that is more than "it just works".

Story break try-it (Robert Recorde) 3 (4 + 4 = 8 and 5 + 3 = 8).

Lesson 3: Let's Plan the Fruit Party

Week 2 (first lesson) • Picture Graphs • Workbook: Lesson 3

Objective	Read and build a picture graph, compare and total data, and judge what a class vote can and cannot tell us.
Thinking skills	Data reading, comparing, sampling awareness, weighing other factors
Materials	Sticky dots, 5 fruit pictures on the wall, counters.
Online set-up	Poll on the slide (thumbs up, no typing needed) and a shared graph the tutor builds live.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"A teacher wants to buy fruit for a class party, but she cannot ask every child one by one. How can we help her? We can make a picture graph."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–7 min Warm-up game and quick questions

DO Quick vote. Each child chooses ONE fruit. Face to face: a sticky dot. Online: when the tutor points to a fruit, give a thumbs up (no typing needed).

SAY *"Everyone chooses ONE fruit. Put your sticky dot on it. (Online: when I point to a fruit, give a thumbs up for yours. No typing needed.)"*

"Look at the wall. Which fruit do you think won? Why do you think so?"

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Which fruit has the most dots?	From the class vote
Which has the fewest?	From the class vote
Are any two fruits tied?	From the class vote
How many votes are there altogether?	Equals the number of children

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): 1 picture = 1 child. Count the pictures: how many children chose blue? (Red has 7 pictures, blue has 5 and green has 3.)	5. Count the blue pictures.
Sneak-peek 2 (thinking): How many MORE children chose red than green?	4 more (7 − 3). Compare the two rows: do not add.

10–25 min Explore (in pairs, with objects)

SAY *"Predict: which fruit will have the most votes and which will have the fewest? Write your guesses down."*

DO 1. Draw a row for each fruit.

2. Put one counter for each child's vote.

3. Write the key: 1 counter = 1 child.

4. Read the graph aloud: most, fewest, same, how many more.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: draw one row for each fruit first, then add the counters.

Confused about the key: "One counter stands for one child. Write it on your graph."

Finished early: ask "How many more votes did the winner get than the fruit in last place?"

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many more votes did the winner get than the fruit in last place?	Take away the smaller from the bigger.
How can you count the total quickly?	Count in 5s, then add the rest.

If everyone has one vote, what is the total?	The number of children.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"How many more chose green than blue? I do not add. I compare. Line up the two rows. Green has 7 and blue has 3. The difference is 7 – 3 = 4."*

"How many children voted altogether? I add all the rows: 5 + 3 + 7 = 15."

"A quick way to count: count in 5s. 5, 10, 15, then three more: 18."

Worked example on the slides	Answer
Reading a graph. How many more chose green than blue? How many children voted altogether?	4 more (7 – 3). Total 15 (5 + 3 + 7).
Count by 5s. A row has 5 faces and another row has 5 faces. Count: 5, 10. Then add the extra faces. Try: 5, 10, 15, then 3 more. How many?	18.

SAY *"The key idea today is: read and compare. The key tells you what one picture means. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Writing 8 + 3 instead of finding the difference (8 – 3).
- Naming only one of two tied fruits.
- Saying "Yes" to the whole school because their class chose it.

35–40 min Brain break

DO Fruit salad. Everyone picks a fruit; when the tutor calls it, those children swap places.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Marie Neurath (Germany, 1898-1986) helped make Isotype, a way of showing numbers with little pictures. More of something means more pictures, all the same size. Link to our lesson: A picture graph uses one small picture for each thing, the same idea Marie used, so we can count the pictures and compare quickly. Then offer the try-it question, only if you like: "Marie draws one small apple picture for each child who chose apples. 6 children chose apples and 4 chose pears. How many apple pictures does she draw? How many more apple pictures than pear pictures?" (the answer is in the answer key). SKIP: if time is short or the group is tired, say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 3 aloud and read questions 1 and 2 aloud. Say “Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder.”

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “What does one picture stand for? Where does it say so?”
- Ask: “Is the question asking how many altogether, or how many more? How can you tell?”
- Ask: “Which row is longest? Which fruit is it? Are two fruits tied?”

IF STUCK Hint 1: “Read the key first.”

Hint 2: “Point to each picture as you count it.”

Hint 3: “Draw two bars side by side and find the difference.”

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children chose blue?	10
Workbook Think harder T2: 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children voted altogether?	24 (12 pictures, each worth 2: count in 2s)
Workbook Think harder T3: 1 picture = 2 children. Six children chose pizza. How many pictures should we draw?	3
Workbook Think harder T4: 1 picture = 2 children. A half picture means 1 child. Apple has 3 pictures and a half. How many children chose apple?	7

Workbook Go further C1: 24 children voted for three fruits: apple, banana and orange. Apple got 6 votes. Orange got 6 more votes than banana. How many votes did banana and orange get?	Banana 6 and orange 12. (24 – 6 = 18 for banana and orange; take away the extra 6: 12, share: 6 each; give orange the 6 back.)
Workbook Go further C2: 1 picture = 2 children. 20 children voted for three fruits. A has 3 pictures. B has 2 more pictures than A. How many children chose C?	4 children (20 ÷ 2 = 10 pictures in all, counting in 2s to 20; A 3 + B 5 = 8, so C has 2 pictures = 4 children).
Workbook Go further C3: Make up your own: 15 children vote for three fruits. Draw a picture graph where banana wins by exactly 2 votes (banana has 2 more than the fruit in second place) and no two fruits are tied. Then try to find a second graph that works.	Open. Check: total 15, banana is ahead of second place by exactly 2, no ties. There are exactly two graphs: banana 7, apple 5, grape 3; or banana 8, apple 6, grape 1.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: "Who would like to share?" Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *"Would the Primary 6 pupils choose the same fruit as us?"*

"What else should the teacher think about before she buys the fruit?"

Discussion question	Listen for
Would the Primary 6 pupils choose the same fruit? What else should the teacher think about before buying?	Not necessarily. A class vote only tells us about our class. The teacher should also think about cost, allergies and how much fruit there is.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Dogs 6, cats 4, fish 3. How many pets altogether?	13

SAY *"A graph tells us about the people who voted. You noticed that, and that is a smart thing to notice."*

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Draw their own picture graph from the Part B tally using one symbol per child.

Answer key (workbook)

Q1 Orange

Q2 Grape

Q3 5 more (8 – 3)

Q4 Banana and watermelon (5 each)

Q5 27 children (6 + 5 + 8 + 3 + 5)

Q6 (a) No. (b) Only our class voted. Other classes may like different fruits.

Q7 Accept sensible ideas: cost, how many are needed, easy to share, keeps fresh, messy to eat, any child who cannot eat it.

Part B Depends on the class vote. Q9: different children like different things, and a small group may not match a whole school.

Think harder T1 to T4 T1: 10 T2: 24 (12 pictures, each worth 2: count in 2s) T3: 3 T4: 7

Think harder slide 1) 6 more. Banana 2, 4, 6, 8, 10 children; orange 2, 4 children; 10 – 4 = 6 (or: 3 extra pictures, count in 2s: 2, 4, 6). 2) Orange 7 (8 + 6 + 4 = 18, count on from 18 to 25: 7). Apple wins with 8. 3) 9. Count in 3s: 3, 6, 9, 12 is 4 pictures, so 1 picture = 3 children. Banana: 3, 6, 9.

Sneak-peek 1 5. Count the blue pictures.

Sneak-peek 2 4 more (7 – 3). Compare the two rows: do not add.

Investigation (What if we change the key?) With 1 picture = 2 children: Cats 2 pictures, Dogs 3, Fish 1. Stays the same: 12 children in all, dogs still win, fish still last, the same number of children chose each. Changes: the number of pictures (it halves). A key of 3 does not work: 4 and 2 are not in the 3s, so we would need part of a picture. A key only works when every count is in its counting pattern (here the

keys 1 and 2 work). Look for: correct rows, the key written on the graph, and a reason that talks about counting in 2s or 3s.

Story break try-it (Marie Neurath) 6 apple pictures. 6 - 4 = 2 more apple pictures.

Lesson 4: Pattern Inquiry

Week 2 (second lesson) • Repeating and Growing Patterns • Workbook: Lesson 4

Objective	Extend a repeating pattern and predict far places using the pattern core; extend a growing stick pattern and work backwards from a total.
Thinking skills	Core identification, prediction by multiples, tables, working backwards
Materials	21 straws, lolly sticks or paper strips per pair (never real matches), shape cut-outs.
Online set-up	Drawing on paper or screen; children use straws, lolly sticks, paper strips or pencils at home (never real matches or toothpicks).
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Patterns are everywhere: in music, in tiles and in the way things grow. Today we will find the rule that makes a pattern repeat, and the rule that makes a pattern grow."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–7 min Warm-up game and quick questions

DO Clap-snap-stomp. The tutor starts a pattern. The class says what comes next, then what comes in the 10th place.

SAY *"I will start a body pattern: clap, snap, stomp. Join in. What comes next? What will we do on the 10th move?"*

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
clap, snap, stomp, clap, snap, stomp. What is the 7th action?	Clap
What is the 8th action?	Snap
What is the 9th action?	Stomp
What is the 10th action?	Clap

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): Beads repeat: red, blue, blue, red, blue, blue … Draw or thread the next 3 beads.	Red, blue, blue. The pattern is red, blue, blue again and again.
Sneak-peek 2 (thinking): Same pattern. What colour is the 12th bead? How do you know?	Blue. The pattern repeats every 3 beads, so the 3rd, 6th, 9th and 12th beads end a repeat, and each repeat ends with blue.

10–25 min Explore (in pairs, with objects)

SAY *"We are building triangles from sticks. One triangle uses 3 sticks. How many sticks do you think 4 triangles will need? Write a guess before you build."*

DO 1. Build 1 triangle with 3 sticks. Count the sticks.

2. Build 2 triangles in a row so they share a side. Count the sticks.

3. Build 3 triangles. Count the sticks.

4. Predict the 4th before you build it.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: build one triangle first and count its 3 sticks.

Confused about sharing: "When you add a triangle, you share one side with the last one. How many NEW sticks?"

Finished early: predict the number of sticks for 10 triangles, then check it with the table.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect

How many sticks for 4 triangles?	9 sticks.
Why do we add only 2 sticks each time?	The new triangle shares a side with the one before.
Can you predict the number of sticks for 10 triangles?	21 sticks.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Red, blue, blue, red, blue, blue. I look for the part that repeats. That part is the core: red, blue, blue. It has 3 shapes."*

"What colour is the 9th? Count in 3s: 3, 6, 9. The 9th ends a core: blue. The 20th: count 3, 6 … 18. The 18th ends a core, so the 19th is red and the 20th is blue."

"For the growing pattern, look at the table: 3, 5, 7, 9. It goes up by 2 each time. Rule: add 2. To work backwards, keep the table going: 3, 5, 7, 9, 11, 13 sticks. 13 sticks make 6 triangles."

Worked example on the slides	Answer
Repeating patterns: find the core. red, blue, blue, red, blue, blue, red, blue, blue … The core is red, blue, blue. It has 3 shapes. What colour is the 9th? The 20th?	9th is blue (3rd of core). Every 3rd bead ends a core: count 3, 6, 9, 12, 15, 18. The 18th ends a core, the 19th is red and the 20th is blue.
Growing patterns: build the table. Add 2 each time. What is the 5th? Work backwards: 13 sticks. How many triangles?	11 sticks. Work backwards with the table: 3, 5, 7, 9, 11, 13 sticks, so 13 sticks make 6 triangles.

SAY *"The key idea today is: core and growth. Repeating: use the length of the core. Growing: find what is added each time. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Saying 3 more sticks each time because one triangle needs 3.
- Thinking the 30th shape is a circle because 30 starts with 3.
- Dividing 15 by 3 to get 5 triangles.

35–40 min Brain break

DO Pattern dance. Make a three-move dance and repeat it. Can the group do the 10th move together?

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Ingrid Daubechies (Belgium, born 1954) liked weaving as a girl. When she could not sleep she doubled numbers in her head: 2, 4, 8, 16 and on. Now she is a maths professor.

Link to our lesson: Doubling is a growing pattern with a rule, and weaving is a repeating pattern, so Ingrid's childhood games are the same kind of pattern the children build today. Then offer the try-it question, only if you like: “Try Ingrid's doubling game: 1, 2, 4, 8, __, __. What is the rule? What are the next two numbers?” (the answer is in the answer key). SKIP: if time is short or the group is tired, say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 4 aloud and read questions 1 and 2 aloud. Say “Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder.”

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Which part keeps repeating? Circle the core.”
- Ask: “How many are added each time? Show me on your sticks.”
- Ask: “How could you find the 20th without drawing all 20?”

IF STUCK Hint 1: “Circle the part that repeats.”

Hint 2: “Count how many are in the core. What is the 3rd shape of the next core?”

Hint 3: “Write the numbers in a table and look at the gaps.”

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: Pattern: circle, triangle, triangle, square, circle, triangle, triangle, square … What is the 14th shape?	triangle
Workbook Think harder T2: Same pattern: how many triangles are in the first 20 shapes?	10. The core circle, triangle, triangle, square has 2 triangles. Count the cores in 4s: 4, 8, 12, 16, 20, so there are 5 cores. Count the triangles in 2s: 2, 4, 6, 8, 10.

Workbook Think harder T3: Red, blue, blue, red, blue, blue … What colour is the 20th bead?	blue
Workbook Think harder T4: Squares in a row need 4, 7, 10, … sticks. With 20 sticks, how many squares can you make? How many sticks are left over?	6 squares and 1 stick left over. The table goes 4, 7, 10, 13, 16, 19; 7 squares would need 22, which is too many; 20 – 19 = 1.
Workbook Go further C1: Beads repeat in a pattern of 5: red, red, blue, blue, green. How many green beads are in the first 47 beads?	9 green beads. Green is the 5th bead of each core, so greens are at 5, 10, 15 … 45: count in 5s, that is 9 numbers. Beads 46 and 47 are red.
Workbook Go further C2: Dots: 1, 3, 6, 10, 15 … What are the next two numbers? What is the 10th number?	21 and 28. The 10th is 55 (add 2, 3, 4, … each time).
Workbook Go further C3: Make up your own repeating pattern with a core of 5 beads (use up to 3 colours). Ask a partner: what colour is the 33rd bead? Write the answer and how you know.	Open. Check: 30 ends a core of 5 (count in 5s), so the 33rd bead is the 3rd bead of the core. Example: red, red, blue, blue, green gives blue.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: "Who would like to share?" Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *"Why do we add only 2 sticks each time, and not 3?"*

"How can we find the 10th without building all 10?"

Discussion question	Listen for
How can we know the 30th shape without drawing all 30?	The new triangle shares a side with the one before, so only 2 new sticks are needed. Start at 3 and add 2 for each new triangle: 3, 5, 7, 9, 11, 13, 15, 17, 19, 21. So 10 triangles need 21 sticks.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Circle, square, circle, square … What is the 9th shape?	Circle (odd places are circles).

SAY *"You can now say what the rule is and use it to find a number far away. That is real maths."*

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Squares in a row (4, 7, 10 ...): find the rule and how many sticks for 10 squares (31).

Answer key (workbook)

Q1 Place 8: triangle. Place 9: square.

Q2 (a) circle (b) square (c) square (d) The pattern repeats every 3 and every 3rd shape is a square. We say 30 when we count in 3s, so the 30th shape ends a core: a square.

Q3 (a) 3, 5, 7, 9, 11, 13, 15, 17, 19, 21 (b) 2 more (c) The new triangle shares a side with the one before. (d) 21 sticks

Q4 7 triangles. Continue the table: 3, 5, 7, 9, 11, 13, 15 sticks.

Think harder T1 to T4 T1: triangle T2: 10. The core circle, triangle, triangle, square has 2 triangles. Count the cores in 4s: 4, 8, 12, 16, 20, so there are 5 cores. Count the triangles in 2s: 2, 4, 6, 8, 10. T3: blue T4: 6 squares and 1 stick left over. The table goes 4, 7, 10, 13, 16, 19; 7 squares would need 22, which is too many; 20 − 19 = 1.

Think harder slide 1) circle. The core has 4 shapes, so count in 4s: 4, 8, 12, 16, 20. The 20th is a triangle, the end of a core; 21st star, 22nd circle, 23rd circle. 2) 9 Bs. Each core A, B, B, C has 2 Bs. 16 letters are 4 cores: 2, 4, 6, 8 Bs. The 17th is A and the 18th is B: 9. 3) Yes. Count in 5s: 5, 10, 15, 20, 25. Each of these ends the core, so the 25th is the last shape of the core, like the 5th.

Sneak-peek 1 Red, blue, blue. The pattern is red, blue, blue again and again.

Sneak-peek 2 Blue. The pattern repeats every 3 beads, so the 3rd, 6th, 9th and 12th beads end a repeat, and each repeat ends with blue.

Investigation (Is it always true?) Yes, always. Any pattern that repeats every 3 shapes has the same shape in places 3, 6, 9, 12 ... 30 (we say these when we count in 3s, so each one ends a repeat). For a

pattern that repeats every 4 shapes, places 4, 8, 12, 16, 20 … match place 4. Look for: a real 3-shape pattern drawn to place 9, testing before agreeing, and the link to counting in 3s (or 4s). Extra: ask "Which place matches place 1 in a 3-shape pattern?" (places 4, 7, 10 … 31).

Story break try-it (Ingrid Daubechies) The rule is double each time (multiply by 2). The next two numbers are 16 and 32.

Lesson 5: Mystery Number

Week 3 (first lesson) • Working Backwards • Workbook: Lesson 5

Objective	Solve "I think of a number" problems by working backwards, find a number machine rule, and solve more-than and fewer-than stories.
Thinking skills	Inverse operations, rule finding, bar-model thinking, logical reasoning
Materials	Paper cups, counters, number machine cards.
Online set-up	Slide with a cup that hides counters; the tutor reveals the total and the children work back.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"I am thinking of a mystery number. I will give you clues, and you have to work backwards to find it. Inquirers love working backwards!"*

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–7 min Warm-up game and quick questions

DO Think-of-a-number magic. The tutor says: "I think of a number, add 3, and the answer is 8." Children work out the number.

SAY *"I think of a number. I add 3. I get 8. What is my number?" Whisper your answer to your partner first."*

"How did you find it? What did you do to the 8?"

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
I think of a number, add 3, get 8. What is it?	5
I think of a number, take away 2, get 6. What is it?	8
I think of a number, add 4, then add 1, get 10. What is it?	5
I think of a number, add 2, then take away 1, get 9. What is it?	8

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): Put 8 counters in a cup. Add 3 more. How many counters are in the cup now?	11. 8 + 3 = 11.
Sneak-peek 2 (thinking): I think of a number. I add 3. Then I add 4. I get 12. What is my number?	5. Undo the last step first: 12 − 4 = 8, then undo the first step: 8 − 3 = 5. Check: 5 + 3 = 8, 8 + 4 = 12.

10–25 min Explore (in pairs, with objects)

SAY *"I hide some counters under the cup and add 5 more. When I tell you the total, can you find how many are under the cup? Guess how you would do it."*

DO 1. Hide a secret number of counters under a cup.

2. Add 5 more counters beside the cup. Tell your partner the total.

3. Your partner works backwards to find the secret number.

4. Lift the cup to check. Then swap roles.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: lift the cup! Check with the counters.

Confused: "What did we do to the secret number? We added. What is the opposite of adding?"

Finished early: make a two-step puzzle for your partner.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How did you find the secret number?	Take away the 5 that were added.

Why does taking away work?	It is the opposite of adding.
What if the story had two steps?	Undo the steps in the reverse order.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Mystery number plus 6 is 14. To find it I go backwards. The opposite of add 6 is take away 6. 14 − 6 = 8. Check: 8 + 6 = 14. ✓"*

"Two steps: I add 5, then I take away 3. I get 9. I must undo the steps in reverse order. The last thing I did was take away 3, so I undo it first: 9 + 3 = 12. Then undo add 5: 12 − 5 = 7. Check: 7 + 5 = 12, 12 − 3 = 9. ✓"

"A number machine: rule add 2. IN 3 gives OUT 5. If OUT is 9, IN was 7."

Worked example on the slides	Answer
Undo it. ? → add 6 → 14 To find ?, go backwards: 14 − 6. What is the mystery number?	8.
Two steps: undo in reverse. I add 5, then I take away 3. I get 9. Go back: undo take away 3 first (9 + 3 = 12), then undo add 5 (12 − 5). What is my number?	7.

SAY *"The key idea today is: the number machine. Number machine rule: add 2. IN 3 gives OUT 5. OUT 9 came from IN 7. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Adding again instead of undoing (12 + 5).
- Two steps: undoing in the wrong order.
- Queue question: giving 16 because Ali is counted twice.

35–40 min Brain break

DO Mystery movers. The tutor whispers a mystery number to one child, who does that many jumps while the rest guess.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: René Descartes (France, 1596-1650) wrote a book in 1637 that used x, y and z for numbers we do not know yet. We still call a mystery number x. Link to our lesson: A mystery number is an unknown number, and Descartes' habit of writing it as x is how we write these problems, then work backwards to find it. Then offer the try-it question, only if you like: "Call the mystery number x. x + 8 = 14. Work backwards: what is x?" (the answer is in the answer key).

SKIP: if time is short or the group is tired, say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 5 aloud and read questions 1 and 2 aloud. Say “Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder.”

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “What did the story do to the number? What is the opposite?”
- Ask: “Which step did you undo first? Why?”
- Ask: “How can you check your answer by going forwards?”

IF STUCK Hint 1: “Draw the story as a picture, a box for the mystery number.”

Hint 2: “Read the last thing that happened and undo it.”

Hint 3: “Try your answer in the story. Does it give the number in the problem?”

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: Ben had some marbles. He gave 5 to Mei, then won 3. Now he has 12. How many did he start with?	14 (12 − 3 + 5)
Workbook Think harder T2: Ben had some stickers. He gave away 6, got 4, then gave away 3. Now he has 10. How many did he have at first?	15 (work backwards: 10 + 3 = 13, 13 − 4 = 9, 9 + 6 = 15; check 15 − 6 + 4 − 3 = 10)
Workbook Think harder T3: A number machine adds the same number each time. IN 4 gives OUT 9. What is the rule? What is OUT when IN is 12? What is IN when OUT is 20?	Add 5. OUT is 17 when IN is 12. IN is 15 when OUT is 20.

Workbook Think harder T4: I think of a number. I add 4, then I take away 7. I get 8. What is my number?	11
Workbook Go further C1: A number machine has two steps: add 3, then add 4. IN 5 gives OUT 12. What goes IN to give OUT 20? Can you make a one-step machine that does the same job?	13 (20 − 4 = 16, 16 − 3 = 13; check 13 + 3 + 4 = 20). A one-step machine that adds 7 does the same job.
Workbook Go further C2: A snail climbs 3 cm each day and slides back 1 cm each night. It starts at the bottom of a 9 cm wall. On which day does it reach the top?	Day 4 (3, 5, 7, then 9 on the fourth day before it can slide back).
Workbook Go further C3: Make up a mystery number story with THREE steps whose answer is 10. Write the story. Show how a partner can work backwards to find 10, and how they can check by going forwards.	Open. Check that the story gives 10 when done forwards and that the partner undoes the steps in the reverse order. Example: I think of a number, add 5, take away 3, add 4, and I get 16 (10 + 5 = 15, 15 − 3 = 12, 12 + 4 = 16).

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: "Who would like to share?" Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY "*Why does taking away work when the story said add?*"

"*What do you do when the story has two steps?*"

Discussion question	Listen for
Ali is 8th from the front and 8th from the back. Why is the answer not 16?	Taking away is the opposite of adding, so it undoes it. With two steps, undo the last step first and go back one step at a time.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
I think of a number. I add 6. I get 15. What is my number?	9

SAY *"Working backwards is a big idea. You explained it in your own words today."*

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Make a three-step mystery number story for a partner.

Answer key (workbook)

Q1 (a) 7 (b) 12 (c) 8 (undo in reverse order: 10 + 2 = 12, then 12 − 4 = 8)

Q2 (a) 15 and 8 (b) Add 4. IN 15 gives OUT 19. OUT 30 comes from IN 26.

Q3 (a) Mei 13 (b) Ben 7 (c) Mei has 6 more than Ben (d) 28 altogether

Q4 (a) 13 children (20 − 7) (b) 5 children (3 + 3 − 1, Ali is counted twice). Talk time: 8th from both ends is 15.

Q5 Open: check that the story has a clear answer.

Think harder T1 to T4 T1: 14 (12 − 3 + 5) T2: 15 (work backwards: 10 + 3 = 13, 13 − 4 = 9, 9 + 6 = 15; check 15 − 6 + 4 − 3 = 10) T3: Add 5. OUT is 17 when IN is 12. IN is 15 when OUT is 20. T4: 11

Think harder slide 1) 10 (Tom is 5th, then 4 children, then Sue: 5 + 4 + 1). 2) 7 (work backwards: 9 + 4 = 13, 13 − 6 = 7; check 7 + 6 − 4 = 9). 3) 11 (undo in reverse order: 14 − 8 = 6, 6 + 5 = 11).

Sneak-peek 1 11. 8 + 3 = 11.

Sneak-peek 2 5. Undo the last step first: 12 − 4 = 8, then undo the first step: 8 − 3 = 5. Check: 5 + 3 = 8, 8 + 4 = 12.

Investigation (Is it always true?) Yes: adding 6 then taking away 6 puts every number back (the steps undo each other). With add 6 then take away 4: 3 gives 5, 10 gives 12, 25 gives 27, so OUT is always 2 more than IN, because the machine really adds 6 − 4 = 2. Look for: four or more tests, a table, and an explanation using "undo" or "6 then back 6". Extra: "Design a machine that always gives 3 less."

Story break try-it (René Descartes) 6 (14 - 8 = 6).

Lesson 6: Coin Combos

Week 3 (second lesson) • Money • Workbook: Lesson 6

Objective	Find all the ways to make an amount, work systematically and use reasoning to rule out cases. (Paying in dollars is an extra on the Think harder page.)
Thinking skills	Systematic listing, comparing, working backwards, justifying "there are no more"
Materials	Play coins (5¢, 10¢, 20¢, 50¢, $1), about 6 of each per pair, play notes ($2, $5, $10, $50), mini whiteboards.
Online set-up	Print the cut-out coins and notes at the back of the workbook, or use real coins and notes at home.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome to the money shop! Today we pay with coins and notes, find every way to make an amount, and make sure we do not miss any and do not count the same way twice."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–7 min Warm-up game and quick questions

DO Coin shop. Three price tags in cents only: eraser 65¢, pencil 80¢, sticker 30¢. Pay exactly with play coins. Any way that pays exactly is a good way. Then look for a way with fewer coins.

SAY *"Here are three price tags in cents: an eraser 65¢, a pencil 80¢ and a sticker 30¢. Pick one and pay exactly with your play coins. Any way that pays exactly is a good way."*

"Now look for a way that uses fewer coins. How many coins did you use? Tell your partner."

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Pay 15¢ with two coins.	10¢ + 5¢
Pay 30¢ with the fewest coins.	20¢ + 10¢
Pay 25¢ with two coins.	20¢ + 5¢
Pay 40¢ with two coins.	20¢ + 20¢

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): Ben has a 20¢ coin, a 10¢ coin and a 5¢ coin. Draw them. Can he pay 25¢ exactly?	Yes: 20¢ + 5¢.
Sneak-peek 2 (thinking): Show one way to make 15¢ with 5¢ and 10¢ coins. Can you find a different way?	10¢ + 5¢, and 5¢ + 5¢ + 5¢. Those are the only two ways.

10–25 min Explore (in pairs, with objects)

SAY *"How many different ways do you think there are to make 20¢ with 20¢, 10¢ and 5¢ coins? Write a guess."*

DO 1. Take 20¢, 10¢ and 5¢ coins.

2. Make 20¢ in as many ways as you can.

3. Record each way on your whiteboard.

4. Ask: have I got them all?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start with the biggest coin that fits, then fill the rest.

Not sure if two ways are different: "Same coins in a different order is the same way."

Finished early: find the ways to make 30¢. How will you know you have them all?

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many ways did you find for 20¢?	4 ways with 20¢, 10¢ and 5¢ coins.

How do you know you have all of them?	Start with the biggest coin and work down.
Are 10 + 5 + 5 and 5 + 10 + 5 different?	No. The same coins, in a different order.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Ways to make 15¢ with 10¢ and 5¢ coins. I start with the biggest coin: one 10¢. Then 5¢ is left, so one 5¢ coin. That is one way: 10 + 5."*

"Now no 10¢ coin. Only 5¢ coins: 5 + 5 + 5. That is two ways altogether."

"25¢ with 20¢, 10¢ and 5¢: with a 20¢, 20 + 5. With 10¢ but no 20¢, 10 + 10 + 5 and 10 + 5 + 5 + 5. With only 5¢, 5 + 5 + 5 + 5 + 5. That is 4 ways. I used a table so I did not repeat any."

Worked example on the slides	Answer
Ways to make 15¢ (coins: 10¢, 5¢). Start with the biggest coin: one 10¢. Then no 10¢: only 5¢ coins.	2 ways.
Ways to make 25¢ (coins: 20¢, 10¢, 5¢). Start with the biggest coin and work down. With one 20¢: 20 + 5. With no 20¢ but 10¢: 10 + 10 + 5, 10 + 5 + 5 + 5. No 10¢ or 20¢: 5 + 5 + 5 + 5 + 5. How many ways altogether?	4 ways.

SAY *"The key idea today is: work in order. Start with the biggest coin and work down. Use a table so you do not repeat a way. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Listing the same way twice in a different order (10+5+5 and 5+10+5).
- Choosing many small coins instead of looking for the fewest.
- Stopping at the first set in Think Harder T1 instead of checking for all of them.

35–40 min Brain break

DO Coin toss count. Toss a pretend coin: heads, 5 star jumps; tails, 10 knee lifts. Count the total together.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Isaac Newton (England, 1642-1727) is famous for science, but he also worked at the Royal Mint, where English coins were made. The Mint cared that every coin had the right weight. Link to our lesson: Coins only work as money when each one is made to the right size and weight, so people can count and trade fairly. Then offer the try-it question, only if you like: "You have one 10-cent coin and some 5-cent coins. You want 20 cents. How many 5-cent coins do you

need?" (the answer is in the answer key). SKIP: if time is short or the group is tired, say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 6 aloud and read questions 1 and 2 aloud. Say "Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder."

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What is your biggest coin in this way?"
- Ask: "Are these two ways the same? Show me with coins."
- Ask: "How do you know there are no more ways?"

IF STUCK Hint 1: "Begin with the biggest coin."

Hint 2: "Keep one coin fixed and change the others."

Hint 3: "Draw a table with a column for each coin."

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: Make 70¢ using exactly 4 coins from 5¢, 10¢, 20¢ and 50¢. Find every set.	2 sets: 50¢ + 10¢ + 5¢ + 5¢, and 20¢ + 20¢ + 20¢ + 10¢.
Workbook Think harder T2: Sam has only 10¢ and 20¢ coins. He pays exactly 90¢ using 6 coins. How many of each coin does he use?	Three 10¢ and three 20¢ (30 + 60 = 90).
Workbook Think harder T3: What is the fewest number of coins to make 45¢?	3 coins (20¢ + 20¢ + 5¢)
Workbook Think harder T4: Find ALL the ways to make 30¢ using 20¢, 10¢ and 5¢ coins. How do you know you have found them all?	6 ways: 20+10; 20+5+5; 10+10+10; 10+10+5+5; 10+5+5+5+5; six 5¢. Start with the biggest coin and work down.

Workbook Think harder T5: Dollars! We also have $1 coins and $2, $5, $10 and $50 notes. (a) A toy costs $17. Pay exactly with the fewest notes and coins. (b) Mei has one $50 note and two $10 notes. A bicycle costs $65 and a helmet costs $8. Does she have enough money for both? Explain.	(a) $10 + $5 + $2 (3 pieces). (b) No. She has $50 + $10 + $10 = $70. The bicycle and helmet cost $65 + $8 = $73. She is $3 short.
Workbook Go further C1: You have one 5¢, one 10¢, one 20¢ and one 50¢ coin. How many different amounts can you make using any of them? What if you also had a $1 coin (100¢)?	15 different amounts: with 3 coins there are 7, and the 50¢ coin doubles that and adds one more. With a $1 coin as well there are 31: each new coin doubles the amounts and adds one more.
Workbook Go further C2: A sweet costs 15¢ and a biscuit costs 20¢. Ben spends exactly 90¢ and buys some of each. How many sweets and biscuits does he buy?	2 sweets and 3 biscuits.
Workbook Go further C3: Make up your own coin puzzle that has exactly ONE answer (for example: "exactly 3 coins that make 35¢"). Write it, swap with a partner and check their answer by listing every possibility.	Open. A good puzzle has exactly one way. Example: 35¢ with exactly 3 coins from 5¢, 10¢, 20¢ and 50¢ has only 20¢ + 10¢ + 5¢. Check by listing every set of 3 coins.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: “Who would like to share?” Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *“How did you make sure you found every way?”*

“Is 10 + 5 + 5 different from 5 + 10 + 5?”

Discussion question	Listen for
Why can’t 35¢ be made with only two coins?	Start with the biggest coin and work down in order. 10 + 5 + 5 and 5 + 10 + 5 use the same coins, so they are one way.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer

Show one way to make 30¢ with 20¢, 10¢ and 5¢ coins.	Any correct way, for example 20¢ + 10¢, or 10¢ + 10¢ + 10¢, or 20¢ + 5¢ + 5¢.

SAY *"A table helped you check. Inquirers check, and today you checked every way."*

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Make 50¢ in as many ways as possible using 5¢, 10¢, 20¢ and 50¢.

Answer key (workbook)

Q1 A: one 20¢. B: two 10¢. C: one 10¢ and two 5¢. D: four 5¢. Row E is spare (only 4 ways). (a) one 20¢ coin (b) four 5¢ coins (c) Start with the biggest coin and work down.

Q2 4 ways to make 25¢: 20+5; 10+10+5; 10+5+5+5; five 5¢. (Worked on the slide.)

Q3 (a) 4 coins: 50¢ + 20¢ + 10¢ + 5¢. (b) e.g. 50¢ + 10¢ + 10¢ + 5¢ + 5¢ + 5¢, or 20¢ + 20¢ + 20¢ + 10¢ + 10¢ + 5¢.

Q4 No. Two coins can total 10, 15, 20, 25, 30, 40, 55, 60, 70 or 100. 35 is not one of them.

Q5 Dollars are now Think harder T5: (a) $10 + $5 + $2 (3 pieces). (b) No. She has $50 + $10 + $10 = $70. The bicycle and helmet cost $65 + $8 = $73. She is $3 short.

Think harder T1 to T5 T1: two sets, 50¢ + 10¢ + 5¢ + 5¢, and 20¢ + 20¢ + 20¢ + 10¢. T2: three 10¢ and three 20¢ (30 + 60 = 90); four coins of 20¢ make only 80¢, so 4 coins cannot make 90¢. T3: 3 coins (20¢ + 20¢ + 5¢); no two coins make 45¢. T4: 6 ways to make 30¢ (20+10; 20+5+5; 10+10+10; 10+10+5+5; 10+5+5+5+5; six 5¢). T5: see Q5 above.

Think harder slide 1) 9 ways (two 20¢: 1 way; one 20¢: 3 ways; no 20¢: 5 ways). 2) 2 ways: $10 + $1 + $1, and $5 + $5 + $2. 3) 4 coins: 50 + 20 + 20 + 5. Three coins cannot work: with a 50¢ coin you need 45¢ from two coins (no pair makes 45¢), and without it three coins make at most 60¢.

Sneak-peek 1 Yes: 20¢ + 5¢.

Sneak-peek 2 10¢ + 5¢, and 5¢ + 5¢ + 5¢. Those are the only two ways.

Investigation (What if we take a coin away?) You cannot make 5¢, 15¢, 25¢, 35¢ or 45¢. You can make 10¢, 20¢, 30¢, 40¢ and 50¢. The amounts you cannot make all end in 5: every coin left ends in 0, so any total ends in 0. Extension: put the 10¢ coin away instead (keep 5¢, 20¢, 50¢): now every amount from 5¢ to 50¢ can be made (e.g. 10¢ = 5¢ + 5¢). Look for: using real coins, a list of what works and what fails, and a reason about last digits (accept "all the coins end in 0").

Story break try-it (Isaac Newton) 2 (10 + 5 + 5 = 20).

Lesson 7: Shape Safari

Week 4 (first lesson) • 2D Shapes • Workbook: Lesson 7

Objective	Name and describe 2D shapes by sides and corners, count shapes in a picture, count triangles in a figure, and compose shapes.
Thinking skills	Classifying, describing, systematic counting, spatial reasoning
Materials	Paper squares, scissors, pattern blocks, 2 shape riddles on cards.
Online set-up	Each child has a paper square to fold and cut on camera; shape hunt at home.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Shapes have secrets. Today we find them by cutting, building and counting. We will count sides and corners, and find shapes hiding inside other shapes."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–7 min Warm-up game and quick questions

DO Shape hunt. Find a circle, a triangle, a square and a rectangle in the room (or at home) in two minutes.

SAY *"Shape hunt! In two minutes find a circle, a triangle, a square and a rectangle in the room (at home for online). Bring one of each, or point to them."*

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
How many sides has a triangle?	3
How many corners has a square?	4
Which shape has no corners?	Circle
What shape is a door?	Rectangle

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): Draw a triangle. Count its straight sides and count its corners.	3 sides and 3 corners.
Sneak-peek 2 (thinking): A shape has 9 corners. How many straight sides does it have? How do you know?	9. A shape has as many straight sides as it has corners.

10–25 min Explore (in pairs, with objects)

SAY *"I will fold this square corner to corner and cut along the fold. What shapes do you think I will get?"*

DO 1. Take a paper square.

2. Fold it corner to corner and cut along the fold.

3. What two shapes do you have?

4. Put them together a different way. Name each new shape.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: line up the long edges so the shapes touch with no gaps.

Not sure of the name: "Count the sides and the corners. What is the name for 4 sides?"

Finished early: can you make a rectangle? A bigger triangle?

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
What two shapes did you cut?	Two triangles of the same size.
What new shapes can you make with them?	Square, bigger triangle, parallelogram.
How many sides does each new shape have?	Square 4, bigger triangle 3, parallelogram 4.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"A triangle has 3 sides and 3 corners. A square has 4 sides and 4 corners. I notice the number of sides is always the same as the number of corners. So an octagon, with 8 sides, has 8 corners."*

"Counting triangles in this picture: I count the small triangles first: 2. Then I look for a big triangle made of more than one small one: 1. Total 3."

"In a counting puzzle we can count a square as a special rectangle. Why? It has 4 square corners, like a rectangle. But on the house picture in Q1, a square goes in the Square column only, so we count each shape once."

Worked example on the slides	Answer
Squares hide inside squares. A 2 by 2 grid. How many squares can you see? Count the small ones first: 4. The whole grid is a square too: 1 more. How many squares altogether?	5 (4 small + 1 big).
A 3 by 3 grid: count by size. Small squares: 9. Squares made of 4 small ones (2 by 2): 4. The whole grid: 1. How many squares altogether?	14 (9 + 4 + 1).

SAY *"The key idea today is: count in order. Count small shapes first, then bigger shapes made of smaller ones. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Counting only the small triangles in the talk-time picture (4 instead of 8): look for bigger triangles made of 2 small ones.
- Counting a square window in both the Square and the Rectangle columns in Q1.
- Thinking a turned or flipped shape is a new one.
- Thinking a square is not a rectangle, or the opposite, without a reason.
- Saying a circle has a straight side (it has 0 straight sides).

35–40 min Brain break

DO Shape statues. The tutor calls a shape; make it with your body or hands. Triangle, square, circle.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Johannes Kepler (Germany, 1571-1630) was an astronomer. In 1611 he noticed that snowflakes have six corners and asked why. Link to our lesson: A snowflake with six corners is shaped like a hexagon, and the children count the sides and corners of shapes today. Then offer the try-it question, only if you like: "Kepler's snowflake has six corners. What shape has 6 straight sides and 6 corners? How many sides do two of these shapes have altogether?" (the

answer is in the answer key). SKIP: if time is short or the group is tired, say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 7 aloud and read questions 1 and 2 aloud. Say “Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder.”

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “How many sides? How many corners? What do you notice?”
- Ask: “Did you count the small triangles first? Now look for the big one.”
- Ask: “Which shape am I? I have 4 sides and all are equal.”

IF STUCK Hint 1: “Trace the shape with your finger and count each side.”

Hint 2: “Colour each small shape a different colour.”

Hint 3: “Count the small shapes first, then the bigger shapes.”

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: Two same-size triangles can make a square. Can they also make a bigger triangle? Draw how.	Yes. Join them along a short side (the two equal short sides touching) to make one bigger triangle. Joining along the long side makes a square instead.
Workbook Think harder T2: How many squares are in a grid 3 squares long and 2 squares high?	8 (6 small squares + 2 squares made of 4 small ones)
Workbook Think harder T3: Three squares are joined in a row to make a long rectangle. How many rectangles can you find altogether? (A square counts as a rectangle.)	6 (3 single squares, 2 pairs and 1 whole row).

Workbook Think harder T4: Draw a shape with 4 straight sides that is NOT a square. Is it a rectangle? How do you know?	Open. A long rectangle has 4 square corners, so it is a rectangle. A slanted shape with 4 sides does not have square corners, so it is not a rectangle.
Workbook Go further C1: A triangle has 4 lines drawn from the top corner to the bottom. How many triangles are there? Follow the pattern 3, 6, 10 … What would 5 lines give?	15 triangles with 4 lines, and 21 with 5 lines (add 2, 3, 4, 5, 6 each time).
Workbook Go further C2: A square is cut into 4 equal squares. One of the small squares is cut into 4 even smaller squares. How many squares can you find altogether?	9 (1 big, 4 medium, 4 tiny).
Workbook Go further C3: Draw your own triangle-counting puzzle: a big triangle with some lines inside. Count the triangles. Swap with a partner. Do you both get the same number? If not, find out who missed some.	Open. Check by counting small triangles first, then bigger ones made of 2, 3 … small ones. Example: one line from the top corner to the bottom gives 3 triangles; two lines give 6.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: "Who would like to share?" Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *"Is a square a rectangle? What makes you say so?"*

"How did you make sure you did not miss any triangles?"

Discussion question	Listen for
How many triangles are in a square with both diagonals drawn (lines joining opposite corners)? How do you make sure you do not miss any?	A square has 4 square corners like a rectangle, so it is a special rectangle. (On the Q1 house picture each shape goes in one column only.) Count small triangles first, then bigger ones made of smaller ones.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer

A shape has 4 sides. Two sides are long and two are short, and all 4 corners are square. What is the shape called? How many corners does it have?	A rectangle. It has 4 corners.

SAY *"You counted in order and explained why a square is also a rectangle. That is careful maths."*

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

How many squares in a grid 4 long and 2 high? (8 + 3 = 11) Then 4 long and 3 high? (12 + 6 + 2 = 20)

Answer key (workbook)

Q1 Circle 1, triangle 2, square 3, rectangle 2. (a) Squares. (b) 8 shapes.

Q2 Straight sides and corners: circle 0 and 0, triangle 3 and 3, square 4 and 4, pentagon 5 and 5, hexagon 6 and 6, octagon 8 and 8. The number of straight sides equals the number of corners. 7 sides means 7 corners.

Q3 (a) square (b) circle (c) triangle

Q4 A: 3 B: 2. (The square with both diagonals, 8 triangles, is the talk-time question: 4 small + 4 made of two small ones.)

Q5 Accept any correct shape with its name: rectangle, square, bigger triangle or parallelogram.

Q6 5 squares (4 small + 1 big)

Think harder T1 to T4 T1: Yes. Join them along a short side (the two equal short sides touching) to make one bigger triangle. Joining along the long side makes a square instead. T2: 8 (6 small squares + 2 squares made of 4 small ones) T3: 6 (3 single squares, 2 pairs and 1 whole row). T4: Open. A long rectangle has 4 square corners, so it is a rectangle. A slanted shape with 4 sides does not have square corners, so it is not a rectangle.

Think harder slide 1) 8 (4 small + 4 made of two small ones) 2) 9: 4 small squares, 2 rectangles 2 long lying down, 2 rectangles 2 long standing up, and the whole grid. 3) Yes to both. The 4 triangles make

the original square again (corners meeting in the middle), and they also fit together into one big triangle whose long side is twice as long as a small triangle's long side (two along the bottom and two on top).

Sneak-peek 1 3 sides and 3 corners.

Sneak-peek 2 9. A shape has as many straight sides as it has corners.

Investigation (Find all the shapes) 3 squares: 2 shapes (a straight line of 3 and a bent shape, like an L). 4 squares: 5 shapes (a line of 4, an L, a T, an S or Z, and a big square). Look for: whole sides touching (not corners only), noticing that a turned shape is the same shape, and a way to know when there are no more (start with a line, then move one square). The 4-square task is the extension: accept 3 or 4 shapes with praise for the method.

Story break try-it (Johannes Kepler) A hexagon. 6 + 6 = 12 sides altogether.

Lesson 8: Logic Lab

Week 4 (second lesson) • Logic and Sorting • Workbook: Lesson 8

Objective	Order things from clues, sort using two rules with a Venn diagram, solve a logic grid, and explain an odd one out.
Thinking skills	Deduction, classification, elimination, justifying
Materials	Two loops of string, buttons or small objects (small objects only with a grown-up nearby), clue cards.
Online set-up	Children sort objects from home by a rule shown on the slide and explain the rule on camera. Buttons and other small objects: only with a grown-up nearby.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Today we are logic inquirers. We will use clues and crosses to find an answer that is hiding. Every cross we write helps us fill in another box."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–7 min Warm-up game and quick questions

DO Guess my number. The tutor thinks of a number from 1 to 20. Ask yes or no questions. How few can we use together?

SAY *"I am thinking of a number from 1 to 20. You may ask yes or no questions. Let us see how few questions we need together."*

"Who would like to share a question they liked? Why was it good? ("Is it more than 10?" cuts the possibilities in half.)"

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Is it more than 10?	A good first question: it halves the numbers.
Is it even?	Another good question.
Which questions are best? Why?	Questions that cross out the most numbers.

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): Ben is shorter than Cara. Cara is shorter than Dev. Draw them in a line. Who is in the middle?	Cara. The order from shortest is Ben, Cara, Dev.
Sneak-peek 2 (thinking): Which number does NOT belong: 4, 8, 12, 15? Give a reason. Can you find a different reason?	15 (it is not in the 4s; it is the only odd number; it is the only one that ends in 5). Accept any reasoned answer.

10–25 min Explore (in pairs, with objects)

SAY *"Here are some buttons. I will sort them into two loops that overlap. Where do you think the buttons that follow both rules will go?"*

DO 1. Lay two string loops so they overlap.

2. Loop 1: buttons with 2 holes. Loop 2: red buttons.

3. Sort all the buttons.

4. Which buttons belong in the middle?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: read one rule at a time. Which buttons follow rule 1?

Not sure where a button goes: "Does it follow BOTH rules? Then it goes in the middle."

Finished early: make a new rule and sort again.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Which buttons went in the middle?	Red buttons with 2 holes.

Which went outside both loops?	Buttons that are not red and do not have 2 holes.
Why does the middle need both rules?	It holds things that follow both rules.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Tia does not drink milk. Sam does not drink milk or juice. I put a cross where it cannot be. Sam cannot have milk or juice, so Sam has water. Now Tia cannot have milk and water is taken, so Tia has juice. Raj has milk."*

"A Venn diagram: numbers 10 to 20. Less than 15 in one loop, even in the other. For each number I ask both questions. 12: less than 15? Yes. Even? Yes. So 12 goes in the middle. 16: less than 15? No. Even? Yes. So 16 goes in the even loop only."

"Every time I write a cross I check whether it gives me a new clue."

Worked example on the slides	Answer
A tick and a cross. Each child drinks a different drink. Tia does not drink milk. Sam does not drink milk or juice. Fill in the grid: put a × where it cannot be.	Sam: water. Tia: juice. Raj: milk.
Venn diagram. Numbers 1 to 10. Rule 1: even. Rule 2: bigger than 5. Middle (both): 6, 8, 10. Only even: 2, 4. Only bigger than 5: 7, 9. Outside: 1, 3, 5. Why does 6 go in the middle?	6 is even AND bigger than 5.

SAY *"The key idea today is: use what you know. Every cross helps you fill in another box. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Putting numbers in the overlap that only fit one rule.
- Giving up on the grid instead of using a cross to rule out.
- Guessing a line-up without checking all the clues.

35–40 min Brain break

DO Odd one out. Show 3 fingers, 3 fingers and 4 fingers. Which is the odd one out? Can you give a second reason?

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Leonhard Euler (Switzerland, 1707-1783) drew circles in his letters to show groups. A small circle inside a big circle is a smaller group inside a bigger one. Link to our lesson:

Euler used circles to show how things sort into groups, which is what the children do in Logic and Sorting. Then offer the try-it question, only if you like: “Draw a big circle for animals and a small circle inside it for cats. Is every cat an animal? Is every animal a cat?” (the answer is in the answer key). Say plainly: the cats and animals picture is our own example, not Euler’s. SKIP: if time is short or the group is tired, say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 8 aloud and read questions 1 and 2 aloud. Say “Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder.”

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Which box can you fill in now? What clue tells you?”
- Ask: “Does this number follow both rules?”
- Ask: “How do you know the answer is the only one?”

IF STUCK Hint 1: “Find a clue that tells you something for sure.”

Hint 2: “Put a cross where it cannot be.”

Hint 3: “Which row or column has only one box left?”

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: In a race, Sam is ahead of Tia. Tia is ahead of Raj. Mei is ahead of Sam. Who is third?	Tia (order: Mei, Sam, Tia, Raj).
Workbook Think harder T2: Ana, Bo and Cy each have a different toy: a ball, a kite or a yo-yo. Bo does not have the kite. Cy has the ball. Who has which toy?	Cy: ball. Bo: yo-yo. Ana: kite.

Workbook Think harder T3: Numbers 1 to 12. Which do we say when counting in 2s? Which when counting in 3s? Which in both?	Counting in 2s: 2, 4, 6, 8, 10, 12. Counting in 3s: 3, 6, 9, 12. Both: 6 and 12.
Workbook Think harder T4: Numbers 1 to 12. Which do we NOT say when counting in 2s or when counting in 3s?	1, 5, 7 and 11
Workbook Think harder T5: Number riddles. Who am I? (a) I am between 20 and 40. When you count in 2s you say me. When you count in 5s you say me too. (b) I have 2 digits. When I add my two digits I get 5. I am less than 20.	(a) 30 (b) 14
Workbook Go further C1: A secret number is between 10 and 30. It is even, and you say it when counting in 3s. It is 2 more than a number you say when counting in 5s. What is it?	12.
Workbook Go further C2: Numbers 1 to 20. Sort them in a two-loop picture with the loops "I say it counting in 2s" and "I say it counting in 3s". How many numbers go in each part? Then use the loops "counting in 2s" and "counting in 4s". What is different, and why?	2s and 3s: 2s only 7, 3s only 3, both 3 (6, 12, 18), neither 7. 2s and 4s: 2s only 5, 4s only 0, both 5 (4, 8, 12, 16, 20), neither 10. Every number we say counting in 4s we also say counting in 2s, so the 4s loop sits completely inside the 2s loop.
Workbook Go further C3: Make up your own "Who has which pet?" puzzle with three friends and three pets. Write clues so that there is only ONE answer. Swap with a partner and check that they find your answer.	Open. Check by trying all 6 ways to hand out the pets. Example: Ana, Bo and Cy have a cat, a dog and a fish. Ana does not have the cat. Bo does not have the dog or the fish. Cy does not have the fish. Answer: Ana fish, Bo cat, Cy dog.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: “Who would like to share?” Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *“Which clue helped you most? Why?”*

“Why does 12 go in the middle of the Venn diagram?”

Discussion question	Listen for
There can be more than one correct odd one out. Can you give a second reason?	12 is less than 15 AND even, so it follows both rules and goes where the two loops overlap.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Ben is shorter than Mei. Mei is shorter than Ali. Who is the tallest?	Ali

SAY *"You used crosses to make a clue, and then a new clue. That is how inquirers solve puzzles."*

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Create a logic grid puzzle with four friends and four pets.

Answer key (workbook)

Q1 (a) Ali tallest, Mei shortest; order Ali, Ben, Mei (b) Cat heaviest, hen lightest (c) Yellow, red, blue, green

Q2 Both (less than 6 and odd): 1, 3, 5. Only less than 6: 2, 4. Only odd: 7, 9. Outside both: 6, 8, 10. (b) 1, 3, 5: they are less than 6 AND odd.

Q3 Ali: fish. Mei: cat. Ben: dog.

Q4 1st Mei, 2nd Ali, 3rd Ben, 4th Sam

Q5 (a) 15. Think harder T5: (a) 30 (b) 14

Q6 Accept any reasoned answer: 11, because we do not say it when counting in 3s (it is also the only 2-digit number); or 6, the only even number.

Think harder T1 to T4 T1: Tia (order: Mei, Sam, Tia, Raj). T2: Cy: ball. Bo: yo-yo. Ana: kite. T3: Counting in 2s: 2, 4, 6, 8, 10, 12. Counting in 3s: 3, 6, 9, 12. Both: 6 and 12. T4: 1, 5, 7 and 11

Think harder slide 1) Cara fish (dog or fish, not dog). Dan: not cat, not rabbit, fish taken, so dog. Ali: not cat, so rabbit. Ben: cat. 2) Eve is first. Ali is 3rd (the middle of 5), Ben 2nd, Cara 5th; 1st and 4th are left for Dan and Eve, and Dan is not first, so Dan 4th and Eve 1st: Eve, Ben, Ali, Dan, Cara. 3) 10, 20 and 30.

Sneak-peek 1 Cara. The order from shortest is Ben, Cara, Dev.

Sneak-peek 2 15 (it is not in the 4s; it is the only odd number; it is the only one that ends in 5). Accept any reasoned answer.

Investigation (Find all the ways) There are 6 orders: Ali Mei Ben, Ali Ben Mei, Mei Ali Ben, Mei Ben Ali, Ben Ali Mei, Ben Mei Ali. With "Ali is not first" 4 are left. Clues that leave exactly one: "Ali is in front of Ben and Ben is in front of Mei" (Ali, Ben, Mei), or "Mei is first and Ali is not last" (Mei, Ali, Ben). Look for: a systematic list (fix who is first, then list the other two) and the reason "each of 3 children can be first and then 2 orders follow". Accept any clue that leaves exactly one order after the child tests it.

Story break try-it (Leonhard Euler) Yes, every cat is an animal. No, not every animal is a cat (a dog is an animal but not a cat).

Lesson 9: Measure It!

Week 5 (first lesson) • Length • Workbook: Lesson 9

Objective	Compare and order lengths, measure with equal units and with a centimetre ruler (also when the object does not start at 0), and change a length from one unit to another in both directions.
Thinking skills	Comparing, measuring, unit reasoning, estimating
Materials	Linking cubes or paper cube strips, paper clips, erasers, objects to measure, a cm ruler for each child, mini whiteboards.
Online set-up	Print the cube strip at the back of the workbook; measure household objects. Each child needs a cm ruler.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"How long is your pencil? "About this long" is not good enough. Today we measure carefully with equal units so everyone gets the same answer."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–7 min Warm-up game and quick questions

DO Longer or shorter? Hold up two objects. Predict which is longer, then lay them side by side to check. Order three objects.

SAY *"I hold up two objects. Predict: which is longer? Now lay them side by side to check. Let me try three objects. Can you put them in order from shortest to longest?"*

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Which is longer, a pencil or an eraser?	Pencil
Put a pen, a pencil and an eraser in order, shortest first.	Eraser, pen or pencil, depending on the objects
How can you check fairly?	Line up the left ends at the same starting line.

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): Ribbon A is 5 squares long. Ribbon B is 3 squares long. Which ribbon is longer? How many squares longer?	Ribbon A, by 2 squares (5 – 3). Draw or count the squares.
Sneak-peek 2 (thinking): Ali measures a table with his pencil and gets 5 pencils. Mei measures the same table with a shorter pencil. Will she count more pencils or fewer pencils? Why?	More. A shorter pencil covers less, so she needs more of them.

10–25 min Explore (in pairs, with objects)

SAY *"Estimate: how many cubes long is your pencil? Write your guess before measuring."*

DO 1. Estimate: how many cubes long is your pencil?

2. Measure with cubes: no gaps, no overlaps, start at zero.

3. Measure a book and a shoe the same way.

4. Then measure the book in hand spans and compare with your partner.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start at the zero end of the object, with no gaps.

Cubes overlap: "Are your cubes touching but not on top of each other?"

Finished early: measure the same object with a different unit and compare the two numbers.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How close was your estimate?	Open answers.

Why must the cubes have no gaps or overlaps?	Equal units laid end to end give a fair count.
Why did you and your partner get different numbers of spans?	Hands are different sizes.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"A pencil is 6 cubes long and a book is 9 cubes long. How many cubes longer is the book? I draw two bars. The difference is 9 – 6 = 3."*

"1 clip is as long as 3 erasers. A pen is 4 clips long. How many erasers? 4 clips and each clip is 3 erasers: 3 + 3 + 3 + 3 = 12 erasers. A bigger unit gives a smaller number, so erasers (shorter) give a bigger number."

"Rules for measuring: equal units, no gaps, no overlaps, start at zero."

Worked example on the slides	Answer
Compare lengths. A pencil is 6 cubes long. A book is 9 cubes long. How many cubes longer is the book? Draw two bars to help.	3 cubes longer (9 – 6).
Bigger unit, smaller count. 1 clip is as long as 2 erasers. A pen is 4 clips long. How many erasers long is the pen? Why is the number of erasers bigger?	8 erasers. Each eraser is shorter, so you need more of them.

SAY *"The key idea today is: equal units. Use equal units, no gaps, no overlaps, and start at zero. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Leaving gaps or overlapping the cubes.
- Not starting at zero.
- Thinking more units means a bigger unit.
- Reading the end mark when the object does not start at 0.

35–40 min Brain break

DO Giant steps. Take your biggest step and your smallest step across the room. Count each: which needs more steps?

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Al-Biruni (Central Asia, 973-1048) loved to measure. He used careful measuring of a hill to work out the size of the whole Earth, and found a value close to today's. Link

to our lesson: Al-Biruni shows that careful measuring of one length can help us find a much bigger one, and the children measure lengths with equal units and a ruler today. Then offer the try-it question, only if you like: “Al-Biruni needed careful measuring. A rod is 10 cm long. A table is 6 rods long. How long is the table in cm?” (the answer is in the answer key). SKIP: if time is short or the group is tired, say “We will keep this story for another day” and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 9 aloud and read questions 1 and 2 aloud. Say “Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder.”

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Where does your object start? Is it at zero?”
- Ask: “Why did you get a different number from your partner?”
- Ask: “Is a bigger unit going to give a bigger number or a smaller number? Why?”

IF STUCK Hint 1: “Line the object up with zero.”

Hint 2: “Draw two bars and mark the difference.”

Hint 3: “Make the story with real cubes.”

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: A ribbon lies from the 3 cm mark to the 11 cm mark on a ruler. How long is it?	8 cm (count the spaces from 3 to 11, or 11 − 3)
Workbook Think harder T2: A red ribbon lies from the 2 cm mark to the 9 cm mark. A blue ribbon lies from the 5 cm mark to the 11 cm mark. Which ribbon is longer? By how many cm?	Red is longer: red 7 cm, blue 6 cm, so red is 1 cm longer.

Workbook Think harder T3: 1 clip is as long as 3 cubes. A pen is 12 cubes long. How many clips long is the pen? A ruler is 5 clips long. How many cubes long is the ruler?	4 clips. 15 cubes.
Workbook Think harder T4: A pencil is 3 erasers long. A ruler is 2 pencils long. A desk is 4 rulers long. How many erasers long is the desk?	24 erasers (a ruler is 3 + 3 = 6 erasers; a desk is 6 + 6 + 6 + 6 = 24)
Workbook Go further C1: A train has 3 carriages. Each is 6 cubes long. There is a gap of 1 cube between carriages. How long is the whole train? What if there are 5 carriages?	20 cubes (18 + 2 gaps). With 5 carriages: 30 + 4 gaps = 34 cubes. There is always 1 gap fewer than carriages.
Workbook Go further C2: A ribbon 30 cubes long is cut into three pieces. The first is 4 cubes longer than the second. The third is 4 cubes shorter than the second. How long is each piece?	14, 10 and 6 cubes.
Workbook Go further C3: Make up your own measuring puzzle that uses TWO different units (for example clips and cubes). Write the puzzle and its answer. Give it to a partner and check their answer.	Open. Check that the link between the units is stated and the answer works. Example: 1 clip is as long as 4 cubes. A snake is 5 clips long. How many cubes long is it? Answer: 20 cubes.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: "Who would like to share?" Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *"Why did you and your partner get different numbers of hand spans?"*

"Why must the cubes have no gaps or overlaps?"

Discussion question	Listen for
Ali measured 6 spans and Mei measured 8 spans for the same desk. Whose hand is bigger? Why?	Hands are different sizes, so hand spans are not equal units. Equal units laid end to end with no gaps or overlaps give a fair count.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer

A ribbon lies from the 3 cm mark to the 9 cm mark on a ruler. How long is it?	6 cm

SAY *"You measured fairly today and you can say why equal units matter."*

88–90 min Look-back (2 minutes)

SAY *"Turn to the Choose and look-back page and look at the box called "Look-back". Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group."*

"Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck."

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later. Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Measure the same object with two different units and find the rule linking the two counts.

Answer key (workbook)

Q1 (a) C (b) B (c) 2 more (d) 12 units (e) Same as A (9 – 2 = 7)

Q2 Open. Check guesses are close. The Difference column is how far off the guess was. Equal units are needed so the measures can be compared fairly.

Q3 Ali. Fewer spans means each span was longer, so Ali's hand is bigger.

Q4 (a) 18 erasers (b) 5 clips

Q5 Ali 11 cubes, Ben 6 cubes. All three together: 25 cubes.

Q6 (a) 12 cm (b) 5 cm (8 – 3) (c) Open: check the child started at the 0 mark and compared with 10 cm.

Think harder T1 to T4 T1: 8 cm (count the spaces from 3 to 11, or 11 – 3) T2: Red is longer: red 7 cm, blue 6 cm, so red is 1 cm longer. T3: 4 clips. 15 cubes. T4: 24 erasers (a ruler is 3 + 3 = 6 erasers; a desk is 6 + 6 + 6 + 6 = 24)

Think harder slide 1) 5 cubes each (5 + 5 + 5 = 15) and 2 cuts: 3 pieces need only 2 cuts. 2) Ben. Six clips of 2 cubes each: 2 + 2 + 2 + 2 + 2 + 2 = 12 cubes. 3) The 11 cm mark (3 + 8 = 11).

Sneak-peek 1 Ribbon A, by 2 squares (5 – 3). Draw or count the squares.

Sneak-peek 2 More. A shorter pencil covers less, so she needs more of them.

Investigation (Is it always true?) Yes, always: more short units are needed to cover the same length. A unit half as long gives double the number (a pencil of 12 cubes is 24 half cubes). Look for: units laid end to end with no gaps, starting at the end of the pencil, both counts written down, and a reason using "shorter unit, more of them". Accept any measured numbers if the shorter unit has the bigger count.

Story break try-it (Al-Biruni) 60 cm (6 x 10 = 60, or 10 + 10 + 10 + 10 + 10 + 10).

Lesson 10: Challenge Day

Week 5 (second lesson) • Mixed Puzzles • Workbook: Lesson 10

Objective	Use patterns, bonds, graphs, money, shapes, logic, working backwards and measuring in new puzzles, and explain the thinking.
Thinking skills	Applying and connecting ideas, perseverance, explaining
Materials	Four puzzle cards (coins, patterns, logic, measuring; printed in the workbook and shown on the Lesson 10 slide), play coins, cubes, certificates, stamps, mini whiteboards.
Online set-up	The four cards are printed in the workbook and shown on the slide. Each pair chooses one card. Pairs swap at minute 20. Reporting back is optional: say one thing you tried, not the answer.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome to challenge day! There are four puzzle cards: coins, patterns, logic and measuring. With every card we use the four inquirer rules: notice, test, check, explain."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–7 min Warm-up game and quick questions

DO Whiteboard round. The tutor asks a quick question from each lesson. Everybody writes the answer on a whiteboard or paper at the same time. No passing round and nobody on the spot.

SAY *"Whiteboard round: I ask a quick question from one of our lessons. Everybody writes the answer on a whiteboard at the same time. There is no passing round and nobody is put on the spot. If your answer is different from your neighbour's, that is a good thing to talk about."*

"It is fine to say "I am not sure". A try-again is always welcome."

DO After the game, ask the quick questions on the warm-up slides (minutes 5 to 7; if time is short, take questions 1 to 3). Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Count in 3s: 3, 6, 9, __	12
10 = 4 + __	6
Red 4, blue 6: how many more are blue?	2
I think of a number, add 4, get 9. What is it?	5
Which coins make 15¢ with two coins?	10¢ + 5¢

7–10 min Sneak-peek puzzles (3 minutes)

SAY *"These two are sneak-peek puzzles. They are ideas we have not learnt yet, so a guess is a great start. "I do not know" is welcome too. Nobody is marked."*

"The first one you can try with counters or a drawing. Write or draw on your own paper. When I say, show it to me only (online: hold your paper up to the camera for me)."

DO Show the sneak-peek slide. Give about a minute for each question, then see the answers without a public board: face to face, walk round and look at papers held low; online, ask children to hold their papers to the camera at the same moment. Do not show or read out anyone's answer. Click to show each answer to the whole group only after all children have had a try, and say "Lots of us had a different idea, and that is fine."

Quietly note who found the second question easy. Do not write scores on the board or say who is "ahead". At minute 40 invite the whole group: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Do not name a child in front of the group. Everyone joins Explore and the Mini-lesson.

At minute 40: read questions 1 and 2 aloud, then pairs read the rest. Say: "When you finish this page you may choose: carry on, or try Think harder." Do not split a pair unless both children agree.

Question	Answer and why
Sneak-peek 1 (any child can try): The pattern is 4, 9, 14, 19 … Write the next two numbers.	24 and 29. The rule is add 5.
Sneak-peek 2 (thinking): Make 35¢ with exactly 3 coins from 5¢, 10¢, 20¢ and 50¢.	20¢ + 10¢ + 5¢. (It is the only way.)

10–25 min Explore (in pairs, with objects)

SAY *"Look at the four puzzle cards. Which do you think will be hardest? Write your guess. Each pair chooses ONE card for 10 minutes, then you swap with another pair. It is fine to report back one thing you tried, not the answer."*

DO 1. Four puzzle cards: coins, patterns, logic, measuring. Pairs get ONE card for 10 minutes.

2. Read the card aloud and test your idea with objects.

3. Then swap cards with another pair for a second 4-minute round.

4. Write your answer and why you know. Online: one card per breakout room; each pair reports in 30 seconds.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: say the inquirer rule that might help.

Rushing: "Check your answer a different way before you move on."

Finished early: make a puzzle for the other pair, then swap.

DO Print the Puzzle cards page from the back of the workbook (or show the Lesson 10 cards slide). Pairs get ONE card for 10 minutes, then swap with another pair for 4 minutes. Answers:

Puzzle card	Answer
Coins: Make 45¢ with exactly 3 coins from 5¢, 10¢, 20¢ and 50¢.	20¢ + 20¢ + 5¢ (the only way).
Patterns: 5, 8, 11, 14 … What is the 10th number? Show how you know.	32. Add 3 each time: 5, 8, 11, 14, 17, 20, 23, 26, 29, 32.
Logic: Mei is taller than Ben. Ben is taller than Sam. Ali is shorter than Sam. Put them in order, tallest first.	Mei, Ben, Sam, Ali.
Measuring: A pencil is 6 cubes long. A pen is 4 cubes longer than the pencil. Put them end to end. How many cubes long is that?	16 cubes (the pen is 10 cubes; 6 + 10 = 16).

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
What did you do first with each card?	Open answers.
Which puzzle was hardest? Why?	Open answers.
What helped you when you were stuck?	An inquirer rule: notice, test, check, explain.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Coins: make 45¢ using exactly 3 coins from 5¢, 10¢, 20¢ and 50¢. I try 20 + 20 + 5 = 45. ✓ Three coins."*

"Pattern: 5, 8, 11, 14 … The rule is add 3. I make a table of places 1 to 10 and keep adding 3: 5, 8, 11, 14, 17, 20, 23, 26, 29, 32. The 10th number is 32."

"Logic: Mei is taller than Ben. Ben is taller than Sam. Ali is shorter than Sam. Order from tallest: Mei, Ben, Sam, Ali. Ali is shortest."

"Measure: a pencil is 6 cubes long. A pen is 4 cubes longer, so the pen is 6 + 4 = 10 cubes. End to end they are 6 + 10 = 16 cubes. (Not 6 + 4: the 4 is only the extra bit.)"

Worked example on the slides	Answer

Cards 1 and 2: coins and patterns. Coins: make 45¢ using exactly 3 coins from 5¢, 10¢, 20¢ and 50¢. Pattern: 5, 8, 11, 14 … What is the 10th number?	Coins: 20¢ + 20¢ + 5¢. Pattern: add 3. Count on: 5, 8, 11, 14, 17, 20, 23, 26, 29, 32, so the 10th is 32.
Cards 3 and 4: logic and measuring. Logic: Mei is taller than Ben. Ben is taller than Sam. Ali is shorter than Sam. Who is the shortest? Measure: a pencil is 6 cubes long. A pen is 4 cubes longer. Put them end to end. How long is that?	Logic: Ali (order Mei, Ben, Sam, Ali). Measure: pen 10 cubes, end to end 16 cubes.

SAY *"The key idea today is: the inquirer rules. Notice the pattern. Test it with objects. Check your answer. Explain why. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Rushing: not checking an answer.
- Not showing how they know.

35–40 min Brain break

DO Freeze dance with maths. When the music stops, the tutor shows a number; make that many with your body.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

DO Story break (optional, 2 minutes): after the movement, show the Story break slide and read it aloud, slowly. In short: Srinivasa Ramanujan (India, 1887-1920) loved numbers and sent his ideas to mathematicians in England. He treated numbers like puzzles to explore. Link to our lesson: Ramanujan treated numbers like puzzles to explore, which is what the children do on challenge day. Then offer the try-it question, only if you like: "Puzzle: find two numbers that add up to 12 and have a difference of 2." (the answer is in the answer key). Say plainly: the taxi 1729 part is a story Hardy told himself. SKIP: if time is short or the group is tired, say "We will keep this story for another day" and go straight to the worksheet. Nothing later depends on it.

40–72 min Worksheet (pairs, then alone, then check together)

DO Minute 40: give out the workbook. Read the title of Lesson 10 aloud and read questions 1 and 2 aloud. Say "Pairs first. Talk before you write. When you finish this page you may choose: carry on, or try Think harder."

About minutes 40 to 54: pairs work together on the first page, with a short teaching slide before it and a check-together slide after it. After that, a short teaching slide, then about 10 minutes (58 to 68) when each child works alone on the second page; the exact times are on each slide. Minutes 68 to 72: everyone stands up and stretches for a minute, then check the second page together. Think harder (T), Go further (C) and the Investigation are not a timed part of the lesson: every child has them in the workbook and may choose them at any time, or at home.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: “Which inquirer rule helps you here?”
- Ask: “How do you know your answer is right? Show me.”
- Ask: “What would you do differently next time?”

IF STUCK Hint 1: “Which lesson does this remind you of?”

Hint 2: “Test it with objects.”

Hint 3: “Explain it to your partner out loud.”

DO Children who choose more: Think harder T1 to T4 (the same four questions are in the workbook and on the Think harder slide), then Go further, then the Investigation on the Choose and look-back page. Never announce who is on which page, and do not split a pair unless both children agree.

Think harder, Go further and Investigation

Question	Answer
Workbook Think harder T1: A pattern starts at 4 and adds 6 each time. What is the 8th number? Is 50 in the pattern?	46 (4, 10, 16, 22, 28, 34, 40, 46). No: the pattern goes 46, 52, so it skips 50.
Workbook Think harder T2: Mei thinks of a number, adds 7, takes away 2 and gets 19. Ben does the same and gets 25. How much bigger is Ben's number?	6 bigger. Work backwards: Mei 19 + 2 = 21, 21 − 7 = 14; Ben 25 + 2 = 27, 27 − 7 = 20; 20 − 14 = 6. (Shortcut: both did the same steps, so the answers differ by the same amount: 25 − 19 = 6.)
Workbook Think harder T3: Ali has 5 coins worth 60¢. He has only 5¢, 10¢ and 20¢ coins. Find every possible set.	Two sets: 20 + 10 + 10 + 10 + 10, and 20 + 20 + 10 + 5 + 5. Make a table by number of 20¢ coins: one 20¢ needs 40¢ from 4 coins (four 10¢); two 20¢ need 20¢ from 3 coins (10 + 5 + 5); three 20¢ need 0¢ from 2 coins (impossible); no 20¢: 5 coins make at most 50¢.
Workbook Think harder T4: Votes: Apple 6, Banana 9, Orange 7. Then 5 more children vote and all choose apple. Which fruit wins now? By how many votes?	Apple wins with 11, which is 2 more than banana (9).
Workbook Go further C1: Triangles in a row use 3, 5, 7 … sticks. Squares in a row use 4, 7, 10 … sticks. Which numbers of sticks, up to 31, can make a row of triangles AND (with the same number of sticks) a row of squares, with none left over? What do you notice?	7, 13, 19, 25 and 31. They go up by 6 (the 2 and the 3 from the two patterns meet every 6). For example 7 sticks make 3 triangles or 2 squares.
Workbook Go further C2: Ali has some stickers. Ben has 4 more than Ali. Mei has 3 fewer than Ben. Together they have 29. How many does each have?	Ali 8, Ben 12, Mei 9.

Workbook Go further C3: Pick any puzzle from today (a puzzle card or a worksheet question). Change ONE thing about it (a number, a rule or a condition) to make a new puzzle. Solve your new puzzle and say what changed in the answer, and why.	Open. Check that the new puzzle can be solved and that the child explains the link between the change and the new answer. Example: change "Make 45¢ with exactly 3 coins" (20¢ + 20¢ + 5¢) to "Make 45¢ with exactly 4 coins": now the only way is 20¢ + 10¢ + 10¢ + 5¢.

These are the questions on the Think harder and Go further pages in the workbook. The Think harder slide shows the same four questions as the workbook (T1 to T4), and the Go further slide shows C1 to C3; their answers are in the answer key below. More are in the separate Question Bank.

72–80 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask for volunteers first: “Who would like to share?” Never call a child by name to answer. Say that passing is always fine. A child may show a drawing or let a partner speak. No tallies, no rankings.

SAY *“Write a puzzle for a friend (workbook Q9) with its answer. Swap with a partner and solve it.”*

“Which puzzle was hardest? Why? What helped you when you were stuck?”

Discussion question	Listen for
Pick one puzzle and teach it to the group in two sentences.	Open answers. Listen for the inquirer rules: noticing a pattern, testing with objects, checking and explaining.

80–88 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
My favourite lesson was ____ because ____.	Open. Use it to plan next steps.

SAY *“You are real inquirers now. Here is your certificate and your completed inquiry journal.”*

“Give each child a certificate. Tell them one specific thing they did well.”

88–90 min Look-back (2 minutes)

SAY *“Turn to the Choose and look-back page and look at the box called “Look-back”. Tick what was hard for you and what helped you. Write or draw one thing you did well. You can pass. If you are still stuck, tick the box. I will look at it after class and we will sort it out next lesson. I will not say your name to the group. This is not shown to the group.”*

“Thank you. Drawing it, using counters, asking a friend or asking for help: those are all good ways to get unstuck.”

DO Show the look-back slide. Keep it to two minutes. Do not ask children to read out their boxes. Never mark or grade the box, and never collect it in front of the group. You may look at it later.

Praise the strategy a child used, not how quickly they finished.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (it is for you only). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson in the open session (a shared breakout room or a hand signal, with a parent nearby).

Extension for children who want more

Write a challenge puzzle for next year's group.

Answer key (workbook)

Q1 23, 28. Add 5.

Q2 5 and 7 (5 + 7 = 12 and 7 – 5 = 2)

Q3 (a) 2 more cats than birds (b) 19 pets (c) Cat and fish (7 + 3 = 10)

Q4 (a) 50¢ (b) 20¢ + 10¢ + 5¢

Q5 (a) square (b) 3 triangles

Q6 Mei is youngest, Ali is oldest.

Q7 5 (5 + 9 = 14, 14 – 4 = 10)

Q8 Pen 10 cubes. End to end 16 cubes.

Q9 Open. Done at talk time: swap puzzles with a partner and solve each other's.

Puzzle cards Coins: 20¢ + 20¢ + 5¢. Patterns: 32. Logic: Mei, Ben, Sam, Ali. Measuring: 16 cubes.

Think harder T1 to T4 T1: 46 (4, 10, 16, 22, 28, 34, 40, 46). No: the pattern goes 46, 52, so it skips 50. T2: 6 bigger. Work backwards: Mei 19 + 2 = 21, 21 – 7 = 14; Ben 25 + 2 = 27, 27 – 7 = 20; 20 – 14 = 6. (Shortcut: both did the same steps, so the answers differ by the same amount: 25 – 19 = 6.) T3: Two sets: 20 + 10 + 10 + 10 + 10, and 20 + 20 + 10 + 5 + 5. Make a table by number of 20¢ coins: one 20¢ needs 40¢ from 4 coins (four 10¢); two 20¢ need 20¢ from 3 coins (10 + 5 + 5); three 20¢ need 0¢ from 2 coins (impossible); no 20¢: 5 coins make at most 50¢. T4: Apple wins with 11, which is 2 more than banana (9).

Think harder slide 1) Two ways: five pieces of 2 cubes; or two pieces of 3 cubes and two pieces of 2 cubes (3 + 3 + 2 + 2 = 10). Three pieces of 3 leave 1 over, and one piece of 3 leaves 7, which 2s cannot fill. 2) Ben, Mei, Ali, Sam. 3) Three 10¢ and four 5¢ (30 + 20 = 50).

Sneak-peek 1 24 and 29. The rule is add 5.

Sneak-peek 2 20¢ + 10¢ + 5¢. (It is the only way.)

Investigation (What if we change the jump?) Totals: jump 1 gives 5 + 6 + 7 = 18; jump 2 gives 21; jump 3 gives 24; jump 4 gives 27; jump 5 gives 30. Stays the same: the first number is 5 and there are always 3 numbers. Changes: the total goes up by 3 each time the jump goes up by 1. Shortcut: the total is 3 times the middle number (3 times 8 = 24), or 15 plus 3 jumps. Look for: a tidy table of jump and total, and any noticing of the +3 pattern. Extra: predict the total for a jump of 10 (45).

Story break try-it (Srinivasa Ramanujan) 7 and 5 (7 + 5 = 12 and 7 - 5 = 2).

Mathematician stories

Each lesson has an optional 2-minute story break after the movement in the brain break (minutes 37 to 39). It is a short, true story about one person who did maths, with one optional try-it question. Children may pass. If time is short, skip it: nothing later depends on it. The answer to each try-it question is in the lesson answer key. Stories where the evidence is thinner are labelled on the slide (Euler: our own cats example; Ramanujan: the taxi story is a story Hardy told himself).

Lesson	Person	Country	Years	Link to the lesson
1	Julia Robinson	USA (born in St Louis, Missouri)	1919-1985	Julia loved counting numbers and arranging objects, and today the children hop along the number strip and find the rule that makes a counting pattern.
2	Robert Recorde	Wales (born in Tenby), worked in England	about 1510-1558	Recorde's equals sign says both sides are worth the same, which is exactly what children check when they balance number bonds like 4 + 4 = 5 + 3.
3	Marie Neurath	Germany (born in Braunschweig); worked in Austria, the Netherlands and England	1898-1986	A picture graph uses one small picture for each thing, the same idea Marie used, so we can count the pictures and compare quickly.
4	Ingrid Daubechies	Belgium (born in Houthalen); now works in the USA	born 1954	Doubling is a growing pattern with a rule, and weaving is a repeating pattern, so Ingrid's childhood games are the same kind of pattern the children build today.
5	René Descartes	France (born in La Haye en Touraine)	1596-1650	A mystery number is an unknown number, and Descartes' habit of writing it as x is how we write these problems, then work backwards to find it.
6	Isaac Newton	England	1642-1727	Coins only work as money when each one is made to the right size and weight, so people can count and trade fairly.
7	Johannes Kepler	Germany (born in Weil der Stadt)	1571-1630	A snowflake with six corners is shaped like a hexagon, and the children count the sides and corners of shapes today.
8	Leonhard Euler	Switzerland	1707-1783	Euler used circles to show how things sort into groups, which is what the children do in Logic and Sorting.

9	Al-Biruni	Khwarazm, Central Asia (in the region of today's Uzbekistan); later worked in Afghanistan and India	973-1048	Al-Biruni shows that careful measuring of one length can help us find a much bigger one, and the children measure lengths with equal units and a ruler today.
10	Srinivasa Ramanujan	India (Tamil Nadu)	1887-1920	Ramanujan treated numbers like puzzles to explore, which is what the children do on challenge day.

Progress tracker

After each exit ticket, write S (secure: correct and can explain), D (developing: correct but cannot explain yet) or H (needs help). Use the notes for what to revisit next lesson.

Student	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Notes

Ending the program

- In Lesson 10, give each child a certificate and their completed inquiry journal.
- Send parents a short message through the approved channel: one thing their child did well, and one thing to keep practising.
- Delete or hand over all Student Data within 7 days after the last lesson (this is in the tutor agreement).

Tutor checklist

Before each lesson

- Read the lesson plan and solve the worksheet yourself without looking at the key.
- Check materials are ready, or the online set-up has been sent.
- Test the platform, screen share and annotation tools 10 minutes early (online).
- Print or open the slide deck for the lesson.

During the lesson

- Circulate and ask questions rather than giving answers.
- Use only the approved channels to contact parents. Speak with children only in the lesson itself, with a parent nearby.
- Do not take photos or recordings unless the Company has approved it and parents have consented.
- If you have a concern about a child's safety or welfare, tell the Company straight away.

After the lesson

- Stamp inquiry journals and fill in the tracker.
- Collect worksheets and answer keys (face to face) and keep them secure.
- Note any child who needs help, and plan a short catch-up at the start of the next lesson.

Math Inquirers Holiday Program

Notice it. Test it. Explain why.

A 10-lesson holiday maths program for Level 1

Your child will spend the holidays as a young mathematician: noticing patterns, testing ideas with real objects and explaining their thinking. Lessons are in small groups, hands-on, and built around thinking, not speed.

What your child will do

- Lesson 1, Count with the Frog: number patterns
- Lesson 2, Number Bond Inquiry: number bonds
- Lesson 3, Let's Plan the Fruit Party: picture graphs
- Lesson 4, Pattern Inquiry: repeating and growing patterns
- Lesson 5, Mystery Number: working backwards
- Lesson 6, Coin Combos: money
- Lesson 7, Shape Safari: 2d shapes
- Lesson 8, Logic Lab: logic and sorting
- Lesson 9, Measure It!: length
- Lesson 10, Challenge Day: mixed puzzles

How you can help

- Ask "How do you know?" when your child tells you an answer.
- Let your child try first. Please sit nearby, but let them find the answer.
- Help gather simple materials: counters or coins, paper and a pencil (online groups).

Practical details

Dates	[start date] to [end date], two lessons a week
Time	[day] and [day], [start time] to [end time]
Where	[venue address] or [online link sent after registration]
Group size	[number] children
Fee	[amount]
Contact	[name, phone or email]

Math Inquirers Holiday Program is an independent enrichment program. It is inspired by enrichment-style thinking skills but is not affiliated with, run by or endorsed by the Ministry of Education or any school.